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Call it Exogenous*

Scaling the Narrative Identification of Fiscal Changes using LLMs

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Abstract We build an LLM-assisted pipeline that transfers the narrative identification framework of Romer and Romer (2010) to developing countries, and we hold it to a robust validation discipline. Country-agnostic codebooks are validated against 44 labelled US acts (1945-2003) through behavioral tests, zero-shot evaluation and manual error analysis, then frozen and deployed to Malaysia (1980–2023), where an agentic identification layer produces an expert-ready inventory of 51 statutory tax, spending and tax-incentive events, each carrying a magnitude, a motivation, a direction and two independent preliminary reads of exogeneity. With no motivation ground truth available, transfer is evidenced three ways: a bilingual corpus that returns comparable inventories in English and Bahasa Malaysia, worked disagreement cases printed in full, and statutory rate paths that reproduce the Vegh and Vuletin (2015) cross-country rate panel exactly wherever the corpus is dense, with every miss predating corpus coverage and every extra an act a top-rate series cannot represent. We demonstrate the dataset with a procyclicality decomposition: Malaysia's statutory-rate cyclicality is statistically indistinguishable from zero, as is the median country's in the same panel, while the narrative labels show that the great majority of Malaysian fiscal actions whose timing looks procyclical carried motivations unrelated to the cycle. We report that as a demonstration and bound it with a label-fragility index rather than a p-value. The exogeneity column is a model read pending expert adjudication, and magnitude and quarterly timing carry no validation record; we state throughout which stages rest on evidence and which do not.

Keywords: Fiscal policy, Narrative analysis, AI for social science

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Note

Draft status. The body, the introduction and the exhibits are in place, including the procyclicality use case in Section 7. Figures whose upstream targets are API-gated and not yet built render as a "pending" note rather than an error.

The Malaysia figures below are built from the deliverable as frozen in June 2026. An evidence-scoping defect found in August 2026 (Section 4.5) has been fixed in the pipeline but the enrichment tails have not yet been re-run, so four of the 51 motivation labels are known to be wrong and are corrected in the narrative, not yet in the exhibits. This now has a quantitative consequence rather than only a cosmetic one: Section 7.4 reports that four relabellings would overturn the decomposition, so the exhibits there should be read as provisional until the tails are re-run.

1 Introduction

Credible fiscal policy analysis needs to distinguish which changes in taxes and spending were responses to the economy and which were not. Romer and Romer (2010) read decades of presidential messages, budget documents and congressional records to recover, act by act, *why* each tax change was legislated, and kept only the acts whose stated motivation was unrelated to current output–calling them exogenous fiscal shocks. The resulting series has been used to show that personal and corporate income tax changes have sharply different macroeconomic effects (Mertens and Ravn 2013), to discipline the anticipation problem that makes fiscal shocks hard even to date (Ramey 2011), and to reconcile narrative and time-series estimates of the tax multiplier that had disagreed for a decade (Mertens and Ravn 2014). The method itself has been shown to travel, at least to another rich country with a comparable archive (Cloyne 2013).

Fiscal policy in developing countries is procyclical: it amplifies booms and busts rather than smoothing them. The stylized fact was named by Kaminsky et al. (2004), and Vegh and Vuletin (2015) sharpen it in the way that matters here, showing that procyclicality holds for statutory tax *policy* and not merely for realized tax revenue, which moves with the cycle mechanically whatever policymakers do. They are also explicit about what their evidence cannot settle. The correlation they document could reflect policy responding to the cycle or the cycle responding to policy, and separating the two requires knowing why each measure was enacted. They name the Romer and Romer (2010) narrative classification as the missing input. That is the gap this paper attacks.

What hangs on it is the credibility of essentially every fiscal policy recommendation made outside the OECD. Fiscal multipliers are not a constant to be borrowed from the United States: they vary with the exchange-rate regime, with openness, and with the level of debt (Ilzetzki et al. 2013), and whether a country escapes procyclicality at all appears to turn on the quality of its institutions rather than on its income (Frankel et al. 2013). For developing Asia specifically, Jha et al. (2014) find that tax cuts do more countercyclical work than spending increases, while conceding that roughly 70% of the revenue variation in their data is non-discretionary.

The method. Narrative identification reads the documentary record of a policy decision to recover the decision's stated motive, and then uses the motive to decide whether the resulting change can be treated as a shock–calling it exogenous. Romer and Romer (2010) and their companion methodological paper (Romer and Romer 2009) apply this to every significant federal tax action in the United States between 1945 and 2007, sorting each into one of four motivation categories and keeping only those whose stated purpose was unrelated to current output. Romer and Romer (2023) later set out the methodological doctrine explicitly, and Section 2 takes those requirements as the specification this pipeline has to satisfy. Two properties make the approach attractive to automate using Large Language Models (LLMs): the evidence for every judgement is a passage another researcher can re-read, and the judgement itself is a reading task rather than an estimation task.

Who has used it. The approach has been replicated and extended into a small industry. Cloyne (2013) built the United Kingdom equivalent from Budget documents and found magnitudes remarkably close to the American ones, which is the single most important precedent for the transfer claim in this paper. Mertens and Ravn (2013) exploited the act-level detail to separate personal from corporate income tax changes. A parallel action-based tradition grew up at the IMF, where Pescatori et al. (2011) coded fiscal consolidation episodes from national policy documents rather than from budget outturns, and Guajardo et al. (2014) used that dataset to show the identification choice is not innocuous: action-based measures overturn the "expansionary austerity" reading that conventional cyclically-adjusted measures produce. Dabla-Norris and Lima (2023) extend the same logic to more than 2,000 consolidation-era tax measures, coding each one's yield, motivation, and date, which is close to the record this paper's pipeline is built to produce.

Where it has been attacked. As one might expect from a subjective classification, the narrative approach is contested, and the contest bears directly on how the results below should be read. Mertens and Ravn (2014) show that structural VARs cannot recover the tax multiplier without assuming the output-elasticity of revenue, that narrative series used as external instruments deliver substantially larger multipliers, and, most relevant here, that measurement error in the narrative series is itself consequential rather than a rounding concern. Hebous and Zimmermann (2018) press harder: the widely-used narrative tax measures correlate only weakly with cyclically-adjusted revenues in both the United States and the United Kingdom, and under weak-instrument-robust inference the implied multipliers are frequently insignificant, so previous literature may understate how uncertain narrative-based estimates are.

The benchmark this paper scores itself against is a hand-built series with acknowledged measurement error, produced by two economists reading documents and exercising judgement. The useful question is therefore not whether an LLM reproduces it exactly, which would be a suspicious result, but whether the disagreements are of the same character as the disagreements two careful human coders would have. Section 5.1 is organised around that question.

Large language models make the reading step cheap, and a small but fast-moving literature has already used them for narrative fiscal identification. Its three leading entries share a design choice.

Each reads a **secondary, harmonised, English-language source**, precisely to avoid the cost of primary national documents. Das et al. (2026) code Economist Intelligence Unit country reports for 64 countries, recovering the direction of spending actions but explicitly not their magnitude. Bhasin and Loungani (2026) read IMF Article IV reports for 17 OECD countries, restricted to austerity episodes. Fritsch and Mazlish (2026) read news wires at daily frequency and do recover magnitudes in percent of GDP, for the United States alone, and for an expectations series rather than a realised one. All three validate the same way, by concordance against a pre-existing narrative or action-based series, which is a strategy that only works where such a series already exists and therefore cannot travel to the countries the method is meant to serve. Table 1 sets the three designs against our own, side by side.

Table 1: The three closest papers, and this one. LLM-based narrative fiscal identification designs compared on the dimensions this paper inverts (Das et al. 2026; Bhasin and Loungani 2026; Fritsch and Mazlish 2026).

Dimension	Das et al.	Bhasin et al.	Fritsch et al.	This paper
Source	EIU country reports	IMF Article IV reports	News wires	National budget speeches, economic reports, plans
Language	English	English	English	English and Bahasa Malaysia
Countries	64	17, all OECD	1, the United States	1 pilot, 4 planned
Instruments	Spending	Austerity episodes	Tax and spending	Statutory tax, spending, incentives
Output unit	Direction of action	Episode	Daily surprise	One row per act and instrument
Magnitude	Not recovered	Percent of GDP	Percent of GDP	Extracted, not validated
Validation	Concordance with benchmark	Concordance with benchmark	Correlation with benchmark	US benchmark, then two no-ground-truth legs

The alternative to concordance validation is visible just outside fiscal policy. Aruoba and Drechsel (n.d.) identify monetary policy shocks from Federal Reserve narrative records and validate not against an existing shock list but against forecast-error predictability and theory-consistent impulse responses. That is the right instinct for a country with no labelled data, and Section 5.2 explains why we cannot yet follow it.

The prior question, whether an LLM can stand in for a human coder at all, has been answered fairly encouragingly. Gilardi et al. (2023) find that zero-shot models beat crowd workers on text annotation by a wide margin and with higher intercoder agreement, and Törnberg (2025) find they beat expert coders and supervised classifiers at multi-country political labelling. Ziems et al. (2024) bound the claim, documenting taxonomic tasks where fine-tuned models still win and concluding that LLMs augment the research pipeline rather than replace it. That is the position this paper occupies. It is also a position Section 5.1.2 gives a mechanism for: the models arrive already knowing the conventional constructs, which is a strength on conventional tasks and a liability on bespoke ones.

Two further literatures constrain the design and are introduced where they bind rather than here. Halterman and Keith (2025) supply the validation framework the paper adopts wholesale (Section 2). Gentzkow et al. (2019) supply the standing objection that fixed text measures notoriously fail to transfer across domains, which is precisely the property our frozen codebooks claim and Section 5.2 has to defend.

We take the opposite side of each trade in Table 1 . This paper builds and validates a pipeline that reads **primary national documents in the local language**, covers the **tax side** as well as spending, returns a **full row per act** (the measure, its direction, its motivation, and a preliminary exogeneity read, each traceable to a quoted passage), and is validated **without assuming ground truth exists**. It is applied end to end to Malaysia, 1980–2023, producing an inventory of 51 announced fiscal events across statutory tax, spending, and tax incentives.

The contribution is a method and a demonstration dataset, not a claim of automation. The pipeline hands a domain expert an audit-ready inventory; the expert still decides what is exogenous. Where our design differs most from a single end-to-end prompt is that it separates each task to provide the most rigurous validation available.

The project this paper reports on is organised around four questions. This paper answers the first two in part and delivers the fourth for one country.

Conjecture 1. *Can an LLM reproduce the narrative identification of fiscal changes for the United States, given the same documents and judged against Romer and Romer's own labels?*

We answer this for three of the five quantities the narrative method recovers: which passages describe a fiscal measure, the motivation behind each act and whether it is exogenous, and the direction of the change. Magnitude in percent of GDP and quarterly timing are not yet built.

Conjecture 2. *Do the same codebooks, frozen, transfer to a country with no labelled data?*

Malaysia is the test. Because no motivation ground truth exists there, the question is answered by consistency evidence, by an external check on the statutory rate paths, and by expert agreement rather than by an accuracy rate.

Conjecture 3. *Can the resulting series support comparable cross-country fiscal multipliers?*

Out of scope here: it needs magnitude and quarterly timing, and more than one country.

Conjecture 4. *Can the pipeline produce a fiscal-change dataset local researchers can actually use?*

The Malaysia inventory is the demonstration, and its provenance chain is the reason it is usable.

Include a brief (1-2 paragraphs) summary of conclusions

The remainder of the paper is organized as follows. Section 2 sets out the two frameworks the pipeline implements: the narrative specification we are automating, and the validation discipline we hold it to. Section 3 describes the pipeline and the dataset it produces. Section 4 reconstructs how the Malaysia dataset was built, including a defect the provenance chain caught after deployment. Section 5 reports the two validation legs, the US benchmark and Malaysia. Section 6 states plainly which stages carry evidence and which do not. Section 7 reports the procyclicality decomposition the dataset was built to make possible.

2 The narrative specification and the validation philosophy

Two ideas carry the rest of the paper. The first is the **specification**: what the narrative method actually asks a reader to extract from a document. The second is the **discipline**: what has to be true before an LLM's output of that extraction can be treated as a measurement. They come from different literatures and they constrain each other, so it is worth stating both before any machinery appears.

2.1 The specification: what narrative identification asks for

This is the specification it hands a coder. Romer and Romer (2023) set out four requirements for narrative analysis to count as evidence rather than anecdote: a reliable source, a clear idea of the information being sought, a dispassionate and consistent approach to the source, and careful documentation of the narrative evidence behind each judgement. Those four requirements are what this pipeline has to satisfy, and they map onto it directly. The reliable source is a corpus of primary government documents. The clear idea of the information sought is a written codebook. The dispassionate and consistent approach is a frozen prompt read at temperature zero, which is consistency of a stronger kind than a human coder can offer. The documentation requirement is met by carrying the verbatim evidence forward with every label, so that each row of the output points back at the sentence that produced it.

Applied to a fiscal act, the method recovers five things: the **measure** (what was changed), the **motivation** (why), the **sign** (whether liabilities rose or fell), the **timing** (when liabilities actually changed, which is not when the law passed), and the **magnitude** (how large the change was). This paper builds the first three and defers the last two, a boundary Section 6 returns to.

Motivation is where the identification lives. Romer and Romer (2010) sort each act into one of four categories according to the reason its own proponents gave for it. Two are responses to the economy: **spending-driven** changes, which finance a decision already taken on the expenditure side, and **countercyclical** changes, which aim to return output growth to normal. Two are not: **deficit-driven** changes, which address a budget deficit inherited from past decisions, and **long-run** changes, which pursue growth, fairness, efficiency, or a different size of government. Only the second pair is exogenous to current output, and only that pair survives into the shock series.

The two boundaries that do the most work are the ones a coder gets wrong. Between countercyclical and long-run, the test is directional: an act meant to return growth to *normal* is countercyclical, while an act meant to raise growth *above* normal is long-run, and an act proposed in normal times for long-run reasons stays long-run even if the economy has weakened by the time it passes. Between spending-driven and deficit-driven, the test is temporal: a tax increase paying for a spending increase enacted within the past year is spending-driven, and the same increase more than a year later is deficit-driven. Das et al. (2026) add a clarifier we adopt verbatim, because it corrects a failure mode LLMs fall into readily: acknowledging current conditions does not by itself make an act endogenous if the stated motive is explicitly non-cyclical.

2.2 The discipline: what makes an LLM output a measurement

The obvious way to scale this is to describe the task to a capable model and read off the answers. Halterman and Keith (2025) show why that is not enough. Their premise is that codebook conceptualisation remains a first-order concern even when the coder is a machine, and their finding is that off-the-shelf models follow real codebooks only imperfectly zero-shot: they attend to label *names* more than to the definitions a researcher wrote under them, they lean on lexical overlap, and they import background concepts a codebook explicitly excludes. An accuracy number computed on top of that behaviour does not establish that the model measured the construct the codebook defines. It may have measured something adjacent that happens to correlate.

Their response is a staged framework, and we adopt it as the paper's validation spine: write the codebook so both humans and models can read it, run behavioural tests that probe whether the model can follow it at all, then evaluate zero-shot against labelled data, then analyse the errors, and only then consider changing the model itself. Each codebook here goes through that sequence, and none of them is trusted before it does.

This project needs the discipline more than most, for a specific reason. The published Romer and Romer act list, and the US documents it was built from, are almost certainly in the pretraining data of any frontier model. Strong agreement on US data is therefore consistent with the model recognising an act by name rather than reasoning from the passage in front of it, and extractable memorisation is known to rise with model size, with duplication in the training corpus, and with the length of the prompt context (Carlini et al. 2022). This is why the behavioural battery includes an explicit memorisation probe, and it is the deeper reason the deployment country carries as much of the argument as the benchmark does: Malaysian budget speeches in Bahasa Malaysia are the part of the evidence a model is least likely to have already seen.

3 The pipeline and what it produces

The method reads primary-source government documents and returns an inventory of fiscal-change events, each carrying a motivation, a sign, and a preliminary exogeneity read. Figure 1 represents the route we adopted to go from one to the other, and it is the map for the rest of the paper: Section 4 walks the route for Malaysia, Section 5.1 reports what the US benchmark certifies about the blue stages, and Section 6 states what the other stages rest on instead.

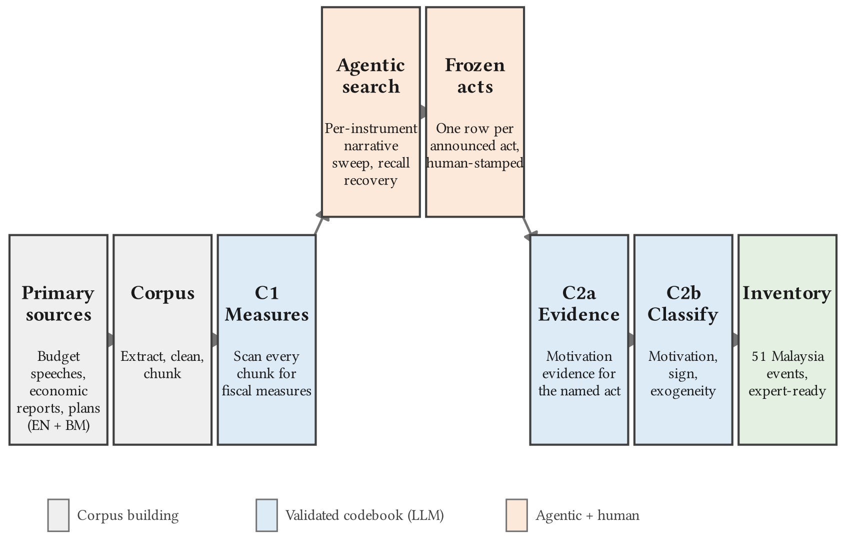

Figure 1: From primary sources to an event inventory. Two LLM codebook stages (blue) are validated against US ground truth; the act-identification step between them (orange) is an agentic narrative search whose output a human stamps before it enters the chain.

Two design commitments distinguish this route from a single end-to-end prompt. The first is that **classification and discovery are separated**. Deciding whether a passage describes a fiscal measure, and why the measure was enacted, is a coding task with a codebook, a gold standard, and a validation record. Deciding which acts exist in a corpus and where each begins and ends is a retrieval-and-consolidation task with no gold standard, and it is handled separately and audited differently. The second is that **every stage is reversible to its source**: each event carries the documents, chunks, and verbatim quotes it was built from, so a reviewer disagreeing with a label can go to the passage rather than to the model.

3.1 The two frameworks the pipeline implements

Section 2 stated the specification and the discipline as ideas; the two tables here state them as work. Table 2 maps the specification: what Romer and Romer (2010) do by hand, and which component does it here. One row carries no step number, because the hand method never has to name it. A human reading a budget speech does not need a rule for deciding that two mentions of a rate cut are the same act; a pipeline does, and Section 4.4 is about how badly that goes when it is automated. Table 3 maps the discipline: the stages a codebook must clear before its output is trusted.

Table 2: The narrative specification. Romer and Romer's (2010) procedure and the component that carries each step here. Step RR5 is where classification credibility lives; act consolidation, which their hand method never has to name as a separate step, is deliberately not automated.

Step	Romer & Romer's step	Our component	Status
RR1	Compile primary-source policy documents	Corpus acquisition and chunking	Built
RR2	Identify significant legislated fiscal changes	C1 measure identification (LLM)	Validated on US
—	Consolidate mentions into distinct acts	Agentic narrative identification	Human-stamped
RR3	Record the sign and size of each change	Sign from C2b; size extracted with the act	Sign validated; size not
RR4	Date the change to the quarter of effect	Announced and effective year	Annual only; C3 not built
RR5	Classify the motivation, filter to exogenous	C2a evidence, C2b classification (LLM)	Validated on US
RR6	Aggregate exogenous acts into a shock series	Signed annual event inventory	Built

Table 3: The validation discipline. Halterman and Keith's (2025) stages and how far each codebook has been taken. S4 is a last resort we have not needed.

Stage	What it establishes	C1	C2
S0	Machine-readable codebook, expert-approved	Complete	Complete
S1	Behavioural tests: legal outputs, memorisation, order	Passed	Passed
S2	Zero-shot evaluation against labelled data	Passed	Passed
S3	Error analysis, ablation, manual adjudication	Passed	Passed
S4	Fine-tuning, if S3 exhausts codebook improvements	Not needed	Not needed

The last row of Table 3 deserves more than a "not needed". Fine-tuning is the obvious lever when a zero-shot codebook underperforms, and we ruled it out before running anything, for reasons that are about the research design rather than about the results.

The first is that fine-tuning and country-agnosticism are in direct conflict. The only labelled data in existence is American, so a fine-tuned model would have been fitted to US legislative vocabulary, US institutions, and US fiscal history. The property the deployment depends on is that the same frozen codebook can be pointed at a Malaysian budget speech and behave the same way, and training on the benchmark is the most reliable way to destroy it. Halterman and Keith (2025) find directly that fine-tuning across codebooks overfits, and Gentzkow et al. (2019) make the more general point that fixed text measures notoriously fail to transfer across domains. The second is arithmetic: 44 labelled acts is two orders of magnitude below the 4,000 to 20,000 examples per dataset that the fine-tuning results in Halterman and Keith (2025) rest on. The third is replication. A frozen prompt is a document another researcher can read, criticise, and re-run; a set of fitted weights is not, and Das et al. (2026) make the same choice for the same reason. The practical constraint, that our access is API-only so fine-tuning would mean moving to a weaker open-weight model, is real but comes fourth.

The consequence is that every improvement in this project had to come from rewriting codebooks rather than from training, which is slower and leaves a visible audit trail of what was changed and why. Section 5.1 reports what that bought, and where it fell short.

3.2 What the pipeline produces

The pipeline covers three instrument families, and the boundaries between them are definitional choices worth stating before any number is read, because they determine what counts as one event.

Statutory tax changes are changes to a standing headline rate, and they split three ways. Corporate income tax tracks the general statutory rate on company profits, with SME, sectoral, and petroleum rates recorded separately. Personal income tax is a progressive schedule rather than a single rate, so the headline series is the top marginal rate and bracket or threshold changes are recorded alongside it. Consumption tax is the broad statutory tax on value added or on sales of goods and services, in whatever local form it takes; introducing such a regime counts as an increase and abolishing it as a cut, which is what makes Malaysia's 2015 GST introduction and its 2018 repeal two events rather than an administrative footnote.

Spending covers discretionary changes: deliberate policy choices such as stimulus and relief packages, subsidy-policy changes, large allocations, Five-Year Plan launches, and consolidation or restraint. It excludes automatic stabilisers and the routine year-on-year drift of the baseline, and it is typed by the component families of Das et al. (2026).

Tax incentives are changes to selective relief: holidays, investment and reinvestment allowances, concessionary investor rates, zones, sectoral and R&D incentives, and personal reliefs.

Incentives and statutory rates both operate on tax liabilities, so they need a rule that keeps them disjoint. Following Klemm and Van Parys (2012), the test is selectivity, temporariness, and mechanism: a measure that is targeted on an investor, sector, region, or activity, or is time-bounded, or works through the base rather than the standing rate, is an incentive, and a universal, permanent change to a standing statutory rate is not. Every event lives in exactly one component, so the three inventories can be read together without double-counting.

Figure 2 represents the whole Malaysia dataset in one frame: 51 narratively-announced fiscal events between 1980 and 2023, split into those three families. Three features of the data are visible before any table is read. Statutory tax changes are dominated by rate *cuts* spread evenly across the period and are mostly read as exogenous. Spending events cluster tightly on 1998, 2009, and 2020–21 and are almost all read as endogenous, which is what a crisis-response spending record should look like. Tax incentives run the full period and are almost uniformly liability-reducing and exogenous, which is what a long-run investment-promotion regime looks like.

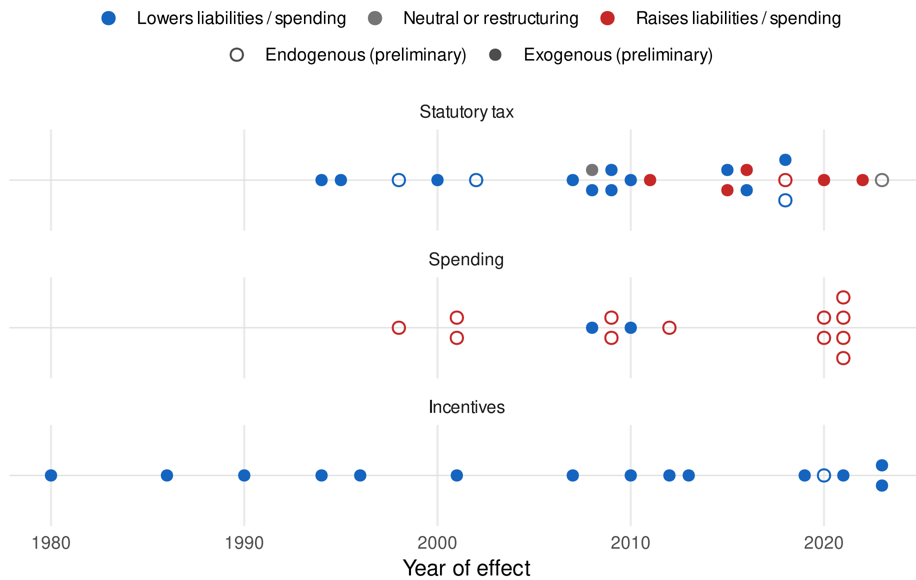

Figure 2: The Malaysia inventory at a glance. One mark per announced fiscal event, positioned at its year of effect; marks stack where several events share a year. Colour is the direction of the change, fill is the preliminary exogeneity read.

Table 4 puts numbers on that picture. Two columns in it are about provenance rather than about fiscal policy: the source documents and evidence chunks count the primary-source passages behind each component, and they are what makes the exogenous share in the last column checkable rather than merely reported. That share is the model's preliminary read, and the spread across components (77% of tax events, 14% of spending events) is doing real work later in the paper.

Table 4: Inventory summary. Coverage and composition per component. Source documents and evidence chunks count the primary-source passages each event was built from; the exogenous share is the model's preliminary read, not an adjudicated finding.

Component	Events	Years	Documents	Chunks	Cuts	Exogenous
Statutory tax	22	1994–2023	41	100	14	17 (77%)
Spending	14	1998–2021	43	109	2	2 (14%)
Incentives	15	1980–2023	23	41	15	14 (93%)
All components	51	1980–2023	84	250	31	33 (65%)

The unit of observation is one narratively-announced act, per instrument. Table 5 sets out the schema. Exhibit 1 below then reproduces one real row in full, which is the faster way to see what a row means.

Table 5: Unit of observation. One row per narratively-announced act × instrument, carrying a magnitude, a motivation label, and a preliminary exogeneity read pending expert adjudication.

Field	Meaning
act	One narratively-announced fiscal act (phased steps collapse to one row)
instrument	CIT / PIT / CONSUMPTION (tax), Expenditure (spending), or incentive category
year	Announced and effective year (quarterly timing is deferred)
magnitude	Δpp statutory rate (tax or rate-bearing incentive) or ringgit amount (spending)
sign	Direction of the effect on fiscal liabilities (increase / decrease / no change)
motivation	Romer–Romer 4-way label from the validated classification codebook
exogenous	Preliminary exogenous/endogenous read — an input for the expert, not a finding
provenance	Source documents, member chunks, and the verbatim quotes behind each field

Exhibit 1 reproduces `MY-CIT-01`, the 1993 announcement cutting Malaysia's corporate rate from 34% to 32% with effect in 1994, and it is worth reading slowly, because every block in it comes from somewhere different and the differences are the point.

The header block is read from the documents: the tax, the announced and effective years, the rate before and after, and the resulting signed effect on liabilities. The **identification** note is the human's, and it records a judgement rather than a fact. This cut was one step of a longer reduction, and the researcher decided to keep it as its own row because the steps were announced in separate budgets rather than legislated once as a schedule. A different, defensible decision would have produced one row instead of several, which is why the note names the decision and the reasoning rather than hiding them. The **motivation** paragraph and the exogeneity verdict come from the classification codebook, reading only the evidence extracted for this act, and the quoted passage below it is that evidence. The **preliminary narrative read** is a second, independent opinion formed during act identification, and printing both is deliberate: where they disagree, the act is flagged rather than resolved. Here the narrative pass could not settle the question and recorded `ambiguous`, so the row goes to the expert with the disagreement visible. Finally, the **sources** are live links to the documents the act was built from.

The general property this illustrates is that no field in the dataset is free-floating. Each is either read from a named primary source, produced by a validated codebook from quoted evidence, or stamped by a human whose reasoning is recorded. A reviewer who disagrees with the label has somewhere to go that is not the model.

Exhibit 1 — Corporate income tax rate reduction 34% to 32% (early-1990s Budget)

Shock ID: MY-CIT-01 | **Tax:** CIT | **Announced:** 1993 | **Effective:** 1994
Rate change: 34% ⟶ 32% (-2 pp; Cut)
Classification (C2b): Exogenous — Long-run
Signed effect on liabilities: decrease (−)

Identification. 34->32 step of the early-1990s reduction; kept as a separate row (researcher decision: separate budget announcements, not one phased act). Evidence: ECON Report 1995 (BM); RMK-7 retrospective path summary.

Motivation (C2b). The evidence consistently frames the corporate tax rate reduction as motivated by structural, long-run objectives: 'enhancing the investment climate', 'improving private sector competitiveness', 'encouraging investments more selectively', and promoting 'technology development and industrial linkages'. These are explicit statements of intent to raise long-run growth and potential output through improved incentives and efficiency, not to offset current cyclical conditions. While strong economic growth is mentioned as context, it is cited as enabling the revenue loss, not as the reason for the action. The stated goals are structural reform and long-run competitiveness, not cyclical stabilisation.

> No explicit motivation quote in surfaced evidence; early-1990s reductions plausibly regional tax-competition context (era).

Preliminary narrative read: ambiguous — left for expert adjudication.

One row in full. Every field below is either read from a named primary source or produced by a validated codebook from the quoted evidence.

Sources: Economic Report / Tinjauan Ekonomi (1995, en); Five-Year Malaysia Plan / Mid-Term Review (1996, en)

3.3 The Malaysia inventory in detail

The exhibits below represent the headline output of the pipeline for Malaysia, 1980–2023, component by component. The three frozen datasets, their data dictionaries, and the full set of worked act write-ups ship with the paper's replication material.

3.3.1 Statutory tax changes

The statutory tax component holds 22 acts: 9 corporate, 9 personal, and 4 on the consumption base. The three tables that follow report them one tax at a time, each row naming the act, the year the change took effect, the change in the headline rate in percentage points, and the motivation and exogeneity the classification codebook assigned.

The corporate record in Table 6 is the cleanest downward path in the dataset: seven cuts between 1994 and 2016 against a single increase, the 2022 Cukai Makmur one-off surcharge. Motivations are almost uniformly long-run, with the model reading the sequence as investment-climate policy rather than demand management. The two departures are informative. The 1998 cut is the one corporate act read as countercyclical, and it lands in the middle of the Asian Financial Crisis. The 2016 cut and the 2022 surcharge are both read as deficit-driven, which is a different exogenous route into the series and one that only appears once fiscal consolidation is on the agenda.

Table 6: Corporate income tax inventory. One row per announced act; Δpp is the headline-rate change; motivation and exogeneity from the validated classification codebook.

Act	Year	Δpp	Motivation	Exogenous
Corporate income tax rate reduction 34% to 32% (early-1990s Budget)	1994	−2	Long-run	Exogenous
Corporate income tax rate reduction 32% to 30% (mid-1990s Budget)	1995	−2	Long-run	Exogenous
Corporate income tax rate reduction 30% to 28% (Budget 1998)	1998	−2	Countercyclical	Endogenous
Corporate income tax rate reduction 28% to 27% (Budget 2007)	2007	−1	Long-run	Exogenous
Corporate income tax rate reduction 27% to 26% (Budget 2007)	2008	−1	Long-run	Exogenous
Single-tier corporate tax system (Budget 2008)	2008	NA	Long-run	Exogenous
Corporate income tax rate reduction 26% to 25% (Budget 2008)	2009	−1	Long-run	Exogenous
Corporate income tax rate reduction 25% to 24% (Budget 2015)	2016	−1	Deficit-driven	Exogenous
Cukai Makmur one-off prosperity tax surcharge (Budget 2022)	2022	9	Deficit-driven	Exogenous

The personal record in Table 7 is less monotone, because a progressive schedule can move in two directions at once. Six of the nine acts cut the top marginal rate, but three raise it, and the recent entries are restructurings rather than simple cuts: the 2016 and 2020 acts add high-earner bands above RM1m and RM2m, and the 2023 act cuts the middle of the schedule while raising its top. Reading the top marginal rate alone would record that last act as roughly neutral, which is why the inventory carries the structural note alongside the rate change.

Table 7: Personal income tax inventory. One row per announced act; Δpp is the change in the top marginal rate, which for restructuring acts understates the change to the schedule.

Act	Year	Δpp	Motivation	Exogenous
Across-the-board individual income tax cut 1pp (Budget 2000)	2000	−1	Long-run	Exogenous
Top individual income tax rate reduction 29% to 28% (Budget 2002)	2002	−1	Countercyclical	Endogenous
Top individual income tax rate reduction 28% to 27% (Budget 2009)	2009	−1	Long-run	Exogenous
Top individual income tax rate reduction 27% to 26% (Budget 2010)	2010	−1	Long-run	Exogenous
Individual income tax restructure, top rate 26% to 25% (Budget 2014, GST-offset)	2015	−1	Long-run	Exogenous
High-earner individual income tax hike 25% to 28% above RM1m (Budget 2016)	2016	3	Deficit-driven	Exogenous
Middle-band individual income tax cut 2pp, RM20k-70k (Budget 2018)	2018	NA	Long-run	Exogenous
New 30% top band above RM2m, 28% to 30% (Budget 2020)	2020	2	Long-run	Exogenous
Individual income tax restructure: M40 cut / T20 hike (Budget 2023)	2023	0	Countercyclical	Endogenous

The consumption record in Table 8 is the shortest and the most eventful, because Malaysia changed regime twice in seven years: the 2015 introduction of a 6% Goods and Services Tax, its 2018 repeal, and the immediate reinstatement of the Sales and Services Tax in its place. The introduction is read as deficit-driven and the repeal as countercyclical, which tracks the documented politics of each.

Table 8: Consumption tax inventory. One row per announced act. The 2018 repeal and the SST reinstatement that replaced it are two rows, since they are two changes to statutory liabilities.

Act	Year	Δpp	Motivation	Exogenous
Service tax rate increase 5% to 6% (Budget 2011)	2011	1	Deficit-driven	Exogenous
Goods and Services Tax (GST) introduction at 6% (Budget 2014)	2015	6	Deficit-driven	Exogenous
GST abolition / zero-rating 6% to 0% (Pakatan Harapan reform, 2018)	2018	−6	Countercyclical	Endogenous
Sales and Service Tax (SST) reintroduction (2018)	2018	NA	Countercyclical	Endogenous

Figure 3 turns the rate-bearing acts back into the statutory rate paths they imply, which is the form a user of the dataset will want: not a list of changes but the level of each headline rate in each year. It also functions as a check, since a path with an implausible jump or a missing step is a sign that an act was missed.

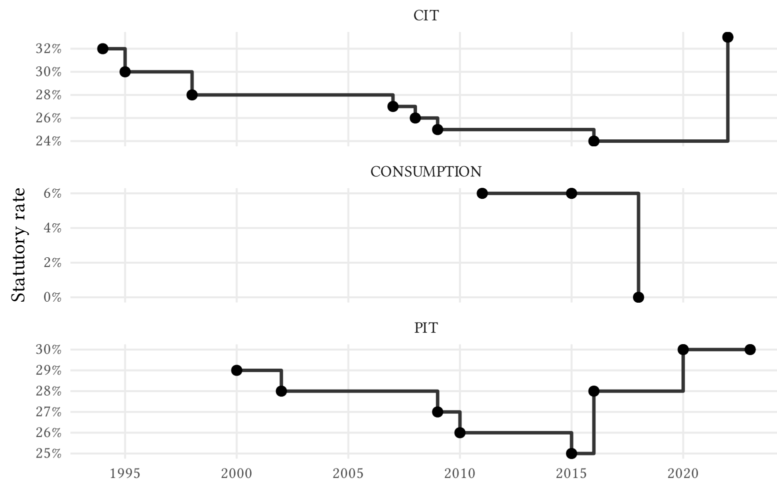

Figure 3: Statutory rate paths by tax type. Step path of the headline statutory rate implied by the rate-bearing acts.

Those paths admit an external check. Vegh and Vuletin (2015) assemble a hand-built statutory-rate panel covering a large set of countries, Malaysia among them, over part of this period. Comparing our reconstructed paths against theirs is the most direct external validation this component admits. Section 5.2.3 runs it: the paths agree exactly wherever the corpus is dense, and every disagreement is either a year the corpus does not cover or an act a top-rate series cannot represent.

The act-level record also carries a distinction an aggregate rate series cannot, and the personal-income acts above are the case in point: several of them move a bracket structure or a base rather than a headline rate. That separation is consequential rather than bookkeeping. Dabla-Norris and Lima (2023), coding more than 2,000 tax measures, find base-broadening markedly less contractionary than rate increases, and Crispolti et al. (2022) show revenue yields differ systematically by instrument and by whether a reform moves the rate or the base. A dataset that recorded only the headline rate would collapse that distinction.

Figure 4 shows when the acts happened and what the model made of them. The visible pattern is that tax policy activity is spread fairly evenly across the period rather than bunched in crises, and that the bars are dominated by the long-run label, with the countercyclical readings concentrated in 1998, 2002, and 2018.

Figure 4: Tax-change timeline. Acts per year × tax type as a diverging stacked bar (hikes up, cuts down); colour denotes the motivation label.

3.3.2 Government spending shocks

The spending component in Table 9 holds 14 acts, and it is the component that looks least like the tax side. Twelve of the fourteen are increases, and the named packages are immediately recognisable as crisis responses: the 1998 and 2001 stimulus packages, the two 2009 packages around the global financial crisis, and the six COVID-era packages from PRIHATIN in 2020 through PEMULIH in 2021. The classification codebook reads twelve of the fourteen as countercyclical and therefore endogenous. Only the two subsidy rationalisations, in 2008 and 2010, come out exogenous, and they come out deficit-driven, which is the reading their documents support.

That 14% exogenous share is not a weakness of the pipeline. A spending record composed largely of endogenous crisis response is what the fiscal history of an emerging market ought to produce, and the fact that the codebook recovers it without being told to is one of the more reassuring results in the paper.

It is worth being explicit that there is nothing to check this component against. The action-based consolidation datasets (Pescatori et al. 2011; Guajardo et al. 2014; Adler et al. 2024) are the benchmark class for narrative spending and consolidation measures, and they are also the series the LLM-based papers validate their own output against. Malaysia is not in them. That absence is simultaneously the reason this pipeline exists and the reason this component reports no concordance statistic. The contrast with Das et al. (2026) is the sharpest available: they reach 64 countries by reading a harmonised secondary source and recover the direction of each spending action without its size, while the ringgit amounts in Table 9 are the trade made in the other direction. Ours are extracted from the announcing document and traceable to it; they are not validated.

Table 9: Government spending-shock inventory. One row per announced spending act; exogeneity from the validated classification codebook, alongside the Das et al. (2026) two-condition preliminary screen.

Act	Effective	Exogenous
National Economic Recovery Plan (NERP) (1998)	1998	Endogenous
Pre-emptive fiscal stimulus package, RM3bn (March 2001)	2001	Endogenous
Additional fiscal stimulus package, RM4.3bn (September 2001)	2001	Endogenous
Fuel-subsidy restructuring (June 2008)	2008	Exogenous
First Economic Stimulus Package, RM7bn (November 2008)	2009	Endogenous
Second Economic Stimulus Package, RM60bn (March 2009 mini-budget)	2009	Endogenous
Subsidy rationalisation programme (2010-)	2010	Exogenous
1Malaysia/Keluarga cash-transfer programme (BR1M -> BKM, 2012-)	2012	Endogenous
PRIHATIN Rakyat economic stimulus package (2020)	2020	Endogenous
PENJANA short-term economic recovery plan (2020)	2020	Endogenous
PERMAI assistance package (2021)	2021	Endogenous
PEMERKASA programme (2021)	2021	Endogenous
PEMERKASA+ package (2021)	2021	Endogenous
PEMULIH package (2021)	2021	Endogenous

Figure 5 makes the clustering visible. Three episodes carry almost the whole component, and the years between them are close to empty, which is the signature of a discretionary spending record rather than a budget series.

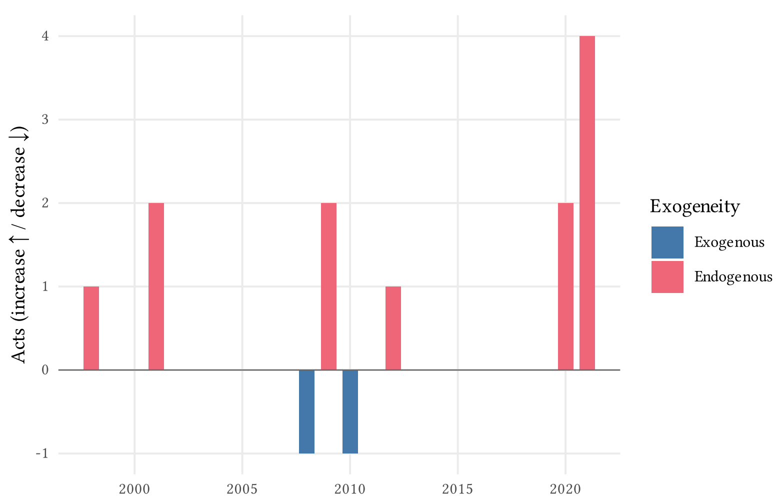

Figure 5: Spending-shock timeline. Acts per year as a diverging stacked bar (increases up, decreases down); colour denotes the exogenous/endogenous read.

3.3.3 Tax incentives and holidays

The incentive component in Table 10 holds 15 acts and is the only one spanning the full period, from the 1980 New Economic Policy equity exemption to the 2023 electric-vehicle package. Every one of the fifteen reduces liabilities, which is what an investment-promotion regime does: incentives are announced and extended far more readily than they are repealed. Fourteen are read as long-run, the exception being the 2020 relocation incentive announced inside the PENJANA recovery package, which the classifier read as countercyclical on the strength of the vehicle it arrived in. That single disagreement is worked through in Section 5.2.

The `Base` column records the tax the incentive operates on. Thirteen sit on corporate income, one on personal income, and one on the consumption base: the 2023 electric-vehicle package works through import-duty, excise, and sales-tax exemptions rather than through a profits rate. That row is a reminder that the component is defined by mechanism, not by base.

The nearest existing comparator is Klemm and Van Parys (2012), who assemble tax holidays, allowances, credits, and corporate rates for more than 40 countries in Latin America, the Caribbean, and Africa between 1985 and 2004. That is an instrument inventory rather than a narrative one: it records what each incentive was, not why it was granted, and it covers neither Malaysia nor this period. The motivation column in Table 10 is what this pipeline adds to that class of dataset, and it is also the column carrying the most interpretive risk, because an incentive's stated rationale is almost always developmental whatever the timing of its announcement. The 2020 relocation incentive is the visible instance. That these measures move real behaviour is not itself in doubt: Agarwal et al. (2017) find temporary consumption-tax holidays produce substantial increases in spending on covered goods that are not offset before or after.

Table 10: Tax-incentive shock inventory. One row per announced incentive change; exogeneity from the validated classification codebook.

Act	Category	Base	Year	Exogenous
DEB equity-restructuring 5pp CIT exemption	Preferential rate	CIT	1980	Exogenous
Promotion of Investments Act 1986 (Pioneer Status / ITA framework)	Tax holiday	CIT	1986	Exogenous
Labuan IOFC / LOBATA 1990 offshore regime	Preferential rate	CIT	1990	Exogenous
Reinvestment Allowance increase (Budget 1994)	Investment allowance	CIT	1994	Exogenous
MSC / Multimedia Super Corridor status incentives	Tax holiday	CIT	1996	Exogenous
Biomass-based generation income-tax exemption (Budget 2001)	Sectoral / R&D	CIT	2001	Exogenous
Iskandar Malaysia (WPI/IDR) zone incentives	Zone incentive	CIT	2007	Exogenous
Green Building Index (GBI) income-tax exemption (Budget 2010)	Sectoral / R&D	CIT	2010	Exogenous
KLIFD / Tun Razak Exchange financial-district incentives (Budget 2012)	Tax holiday	CIT	2012	Exogenous
Angel-investor & venture-capital tax incentives	Other	PIT	2013	Exogenous
Principal Hub 10% concessionary rate (Budget 2019)	Preferential rate	CIT	2019	Exogenous
Companies-relocating-to-Malaysia incentive (PENJANA / Budget 2021)	Preferential rate	CIT	2020	Endogenous
Pharmaceutical/vaccine 0-10% preferential rate (Budget 2021)	Preferential rate	CIT	2021	Exogenous
Relocation incentive extension (Budget 2023)	Preferential rate	CIT	2023	Exogenous
Electric Vehicle (EV) tax incentives (Budget 2023)	Sectoral / R&D	Consumption	2023	Exogenous

Figure 6 shows the resulting timeline: a steady drip of liability-reducing measures across four decades with no repeals, which is a different fiscal object from either of the other two components and the reason they are kept apart.

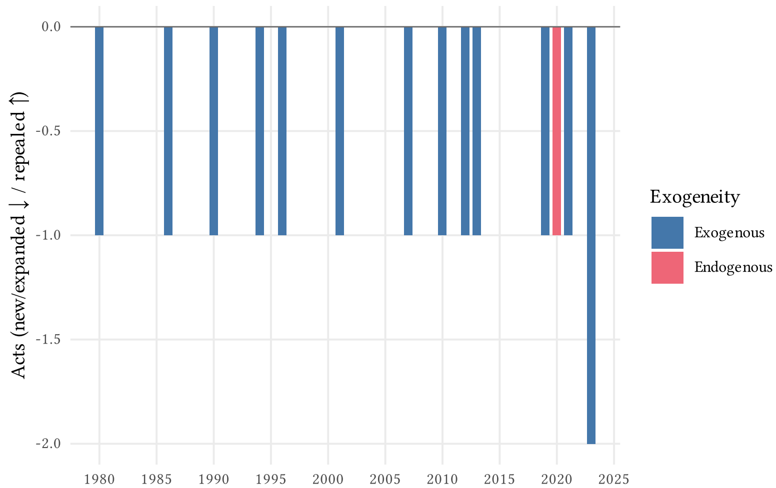

Figure 6: Incentive-shock timeline. Acts per year as a diverging stacked bar (new or expanded down, repeals up); colour denotes the exogenous/endogenous read.

4 How the Malaysia dataset was built

This section walks Figure 1 for Malaysia, stage by stage, so the dataset's provenance is reconstructible. The short version: a 168-document primary-source corpus was scanned exhaustively by a validated codebook, the scan was used as a seed rather than as the event list, acts were consolidated by an agentic narrative search that was audited for recall and stamped by a human, and only then were the validated motivation codebooks run against each act's own evidence.

4.1 The corpus

Malaysia's fiscal record is published across four document series, none of which alone covers the period. Budget Speeches announce measures but do not always report the rates they replace; Economic Reports carry the retrospective rate tables; Bank Negara annual reports carry the macro framing that motivation classification depends on; and the Five-Year Plans carry the structural and development-expenditure commitments. The corpus takes all four, plus the standalone crisis packages. It is also bilingual: ten Economic Reports were published in both English and Bahasa Malaysia, and that parallel decade is what makes the cross-language test in Section 5.2 possible.

Table 11 reports what was actually read. The distinction the `Pending` column draws is important and easy to lose: the acquisition pass catalogued 168 documents, but 55 of them sit in archives we have not yet obtained, so the corpus the codebooks actually read is 113 documents and 22,050 extracted pages. Counting the catalogue rather than the corpus would overstate coverage by a third.

Table 11: The deployment corpus. Primary-source document series behind the Malaysia dataset. Documents, pages and year span describe the extracted corpus; `Pending` counts documents catalogued by the acquisition pass whose PDFs have not yet been obtained and which therefore contributed nothing.

Series	Language	Documents	Pending	Pages	Year span
Bank Negara Malaysia Annual Report	English	27	17	6021	1997–2023
Budget Speech / Ucapan Bajet	English	25	19	1915	1999–2023
Crisis booklet / fiscal stimulus package	English	8	0	435	1998–2021
Economic Report / Tinjauan Ekonomi	Bahasa Malaysia	10	0	1736	2014–2023
Economic Report / Tinjauan Ekonomi	English	29	15	5432	1995–2023
Five-Year Malaysia Plan / Mid-Term Review	English	14	4	6511	1981–2023
All series	—	113	55	22050	1981–2023

Figure 7 shows how those pages are distributed over time, and it carries a caveat the totals hide. The deep archive is recent. Before the mid-1990s only the Five-Year Plans are available to us, and the annual series begin in 1995 (Economic Reports), 1997 (Bank Negara) and 1999 (Budget Speeches). The dataset nonetheless contains events back to 1980, because the Economic Reports and Plan documents carry retrospective rate tables and historical narrative. That is a legitimate route to an early event, but it is a weaker one: an act recovered from a later document's summary table has less motivation evidence attached than an act read from the speech that announced it. Early-period rows should be read with that in mind.

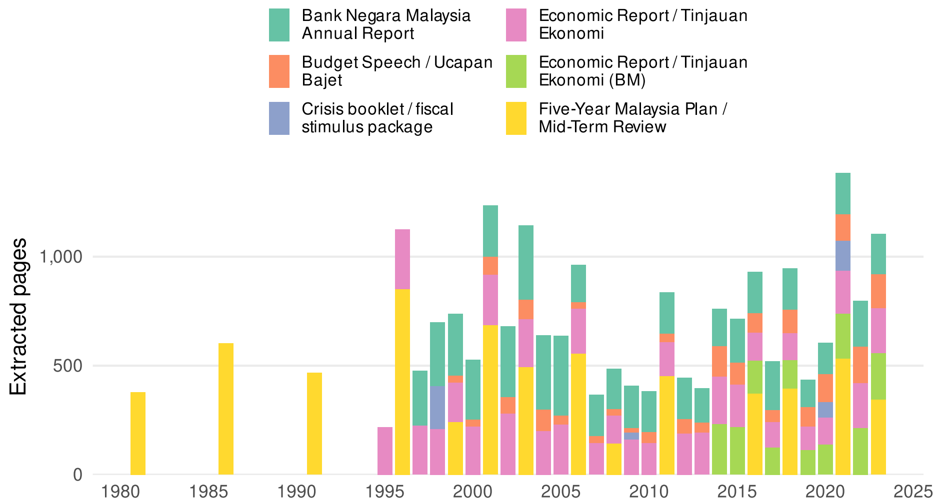

Figure 7: Corpus coverage by year and series. Extracted pages available in each year, stacked by document series. The bilingual Economic Reports (BM) mark the decade the cross-language consistency test uses.

Extraction and cleaning produce a page-level text stream, which is then segmented into the unit the measure-identification codebook actually reads: 3,194 chunks across 113 documents. The segmentation is a sliding window over pages rather than a token splitter: ten pages per chunk, advancing seven pages at a time, so consecutive chunks overlap by three pages. Pages are joined with an explicit page-break marker, chunks shorter than 100 characters are dropped as extraction artifacts, and a 40,000-token ceiling raises a warning rather than splitting a chunk.

The choice of a page window over a semantic or token-based splitter is deliberate. A budget speech states a measure in one sentence and its justification several paragraphs later, sometimes on the next page; a splitter that cuts on token count would routinely separate the two, and the motivation stage would then be asked to classify a measure whose stated reason is in a different chunk. The overlap exists for the same reason, so that a measure near a page boundary appears whole in at least one chunk. The cost is redundancy, since the same passage is read more than once, and that cost is paid at the identification stage rather than passed to the reader.

4.2 The codebook scan and what it is for

The measure-identification codebook reads every chunk and returns the fiscal measures discussed in it. On the Malaysia corpus it surfaced **7,676 measure-mentions** across 1,193 chunks, at a median of five measures per chunk and a maximum of 42 — a density that matters twice below, once here and once in Section 4.5.

That scan is a *seed*, not the event list, for two reasons. The first is mechanical: the handoff to the motivation stage keeps only the highest-ranked measure per chunk, which discards **84.5%** of the mentions and leaves 1,036 measure rows. The second is conceptual: a scan returns mentions, and one act generates many mentions across many documents and both languages, while some acts generate mentions the scan phrases differently each time. Turning mentions into acts is a separate problem, and it is the subject of the next two subsections.

4.3 Agentic narrative identification

Acts are identified per instrument by an agentic procedure that reads the corpus directly, in seven steps.

1. **Instrument configuration.** The concept is defined (statutory corporate income tax; broad consumption tax; discretionary spending; investment incentives), together with the surface terms it takes in the corpus, in both English and Bahasa Malaysia, and the document series that anchor it.
2. **Keyword scan.** The configured terms are matched against the codebook's measure pool. This is the recall floor and it is logged as a count, with false positives named.
3. **Semantic sweep.** The matched measures and their chunk text are read, and the search broadens past the keyword list to catch events phrased without the vocabulary — a rate change written as "reduce the rate to 24 per cent" and never as "corporate income tax".
4. **Recall recovery.** For rate instruments, the implied statutory rate path is reconstructed and checked for missing steps; for spending and incentives, the candidate set is checked against pre-declared known-act checkpoints. Where a step is missing, the anchor document's full text is read directly, behind the chunked view. This step is what recovered Malaysia's 2016 corporate rate cut and its 2022 Cukai Makmur surcharge, neither of which the scan surfaced.
5. **Consolidation.** Surface forms and recovered events are grouped into announced acts. A phased multi-step cut is one row carrying a schedule, not four rows; an omnibus act touching two taxes is two rows, one per instrument. A preliminary exogeneity read is recorded, anchored in a verbatim quote.
6. **Human review.** The consolidated events, the recall audit, and every open adjudication question are presented for review. Nothing is written until the researcher confirms the events, the consolidation grain, and the preliminary reads.
7. **Freeze.** The stamped dataset is written with a provenance record naming the reviewer, the commit, and the date, and a notebook that renders its contents back from the frozen file so the write-up cannot drift from the artifact.

The agentic step earns its place at step 4. The codebook scan is a strong first pass, and better than the design expected. Although it is scoped to tax liabilities, it surfaced 189 spending measures and 151 incentive measures, because Malaysian budget and central-bank documents discuss stimulus, subsidy, and incentive policy in the same breath as tax. On the track it should have been best at, though, the statutory rate path, it was not sufficient on its own. Reconstructing the corporate rate path and asking which mentions account for each step exposed gaps that no amount of reading the scan's output would have revealed, and reading the anchor documents directly, behind the chunked view, recovered them. Malaysia's 2016 corporate rate cut and its 2022 Cukai Makmur surcharge both entered the dataset that way.

That is a specific claim about division of labour, not a claim that the codebook underperformed. A chunk-level classifier answers "does this passage discuss a fiscal measure", which is the question it was validated on. It was never asked "is the rate path complete", because a chunk cannot answer that: completeness is a property of the corpus, not of any passage in it. The agentic pass is what turns the second question into something answerable, by holding a model of the instrument and checking the corpus against it.

Recall is nonetheless the property this dataset can least defend on its own. Precision is checkable by anyone holding it, since each row names its sources and quotes. Recall is not: no reader can see what was never found. The recall scorecards frozen with each component (reproduced in Section A) record what was searched for and what the search returned, and they are useful as a transparency record, but they are not a recall measurement. They are the search describing itself, and a search cannot certify its own coverage.

The honest assessment has to come from outside, and it is the assessment currently being collected: domain experts on Malaysian fiscal history are being asked which significant acts are missing from the inventory, which is a question only someone holding an independent mental list can answer. That procedure is subjective, and we would rather say so than substitute a number that looks objective and measures the wrong thing. Section 5.2 sets out the design; its results are not yet in.

4.4 Act consolidation without automated record linkage

We built an automated act-aggregation step for this purpose and did not use it. The step clusters the scan's surface forms into canonical acts and was scored against the 49 Romer–Romer reference acts, where it recovers most of them. On the Malaysia pool of 877 distinct surface forms it fails in the way that matters for an event layer: it fragments the instrument tracks that need to stay whole — the corporate-rate track shattered into near-singletons, the consumption-tax track split across nine clusters, and it never bridged the English and Bahasa Malaysia names of the same act. Both failures are recorded in the frozen recall audits in Section A.

What we ran into is the oldest problem in record linkage, and it helps to name it as such. Deciding whether two mentions refer to the same underlying entity is the decision problem Fellegi and Sunter (1969) formalised, and the modern taxonomy of approaches to it is surveyed by Binette and Steorts (2022). Every method in that literature trades off the same two errors we hit: merging records that belong apart, and splitting records that belong together. Economics has shown the trade-off can be managed well at scale, with Abramitzky et al. (2021) documenting automated historical linkage whose false-positive and match rates stand comparison with hand linkage. Two features of our problem sit badly with those results. Their linkage keys are person-level attributes with stable distributions, whereas an act's surface form is a phrase a speechwriter chose. And their algorithms are tuned and audited against hand-linked training pairs, which for Malaysian acts do not exist. The conventional finding, that record linkage automates well, is therefore compatible with ours, that this instance of it does not.

The finding is a negative result about clustering, not about LLMs: consolidating mentions into acts requires knowing which mentions describe the same *statutory instrument*, and surface-form similarity does not carry that. The agentic procedure of Section 4.3 succeeds where clustering fails because it works from the instrument down, reconstructing the rate path and then asking which mentions account for each step, rather than from the mentions up. That is also why it needs a human stamp: the reconstruction is a judgement about what counts as one act, and the researcher owns it. Keeping a human in that loop is the conventional position in the linkage literature rather than a concession, since the algorithm-versus-human comparisons that would license full automation are exactly what a country with no hand-linked acts cannot run.

4.5 A defect the evidence chain surfaced

The motivation stage is two codebooks: an extraction step that pulls the motivation evidence for an act out of a passage, and a classification step that assigns the four-way label from that evidence. During reviewer inspection of two corporate-tax rows in August 2026, both codebooks turned out to be reading the wrong act's evidence, through two independent mechanisms.

The first was a join. Evidence had been extracted once per chunk, under whichever measure the scan ranked first — the 84.5% discard of Section 4.2 — and events were matched to that store on chunk identity alone. But the identification procedure cites a chunk because the act's sentence is in it, not because the act was ranked first. 95 of 100 tax member-chunk rows, and every spending and incentive row, were structurally exposed. The second was the extraction codebook itself, which had been instructed to extract *all* motivation evidence in a chunk. That is harmless on the US development chunks, which discuss one act each, and wrong on Malaysian budget speeches carrying a median of five measures and a maximum of 42.

An audit of the 14 events where the model and the identification pass disagreed about exogeneity found four classifications built on another act's evidence: two corporate-tax rows, two personal-income-tax rows. Both mechanisms are fixed — extraction is now scoped to the named act, and evidence is extracted per event rather than per chunk — and the enrichment has not yet been re-run, so the four rows remain wrong in the exhibits above.

We report this as a result rather than as a maintenance note, because it is the kind of failure the validation framework exists to catch and did not. Both codebooks passed every US test. They failed on a distributional difference between the validation corpus and the deployment corpus — single-act chunks versus multi-measure budget speeches — that no US test could have exposed. The cost is quantified because the provenance chain made it quantifiable: four wrong labels out of 51, found by following each label back to the passage it came from. A pipeline without that chain would have shipped them.

5 Validation

The pipeline described so far produces an inventory. This section asks whether that inventory should be believed.

The **US benchmark** (Section 5.1) answers the first question: given the same documents Romer and Romer read, and judged against the labels they published, can an LLM reproduce their identification? That question has a correct answer to score against, so it is answered with accuracy statistics.

Malaysia (Section 5.2) answers the second: do the same codebooks, frozen, still work in a country with no labelled data? That question has no correct answer to score against, so it is answered by consistency evidence and expert agreement instead. The second is the harder question and the more important one.

5.1 The US benchmark

Can an LLM, given the same primary documents and judged against Romer and Romer's own labels, reproduce their identification of US fiscal changes?

The ground truth we score against needs stating plainly, because everything in this section depends on what it is and is not. Romer and Romer (2010) published, alongside their results, an act-level record, since archived as replication data (Romer and Romer 2019): for each of the significant federal tax actions between 1945 and 2007, the act's name, its motivation category, and the narrative reasoning behind that classification, set out at length in their companion narrative analysis (Romer and Romer 2009). After matching that record to the US source documents our pipeline reads, 44 acts carry both a published label and locatable passages in the corpus. Those 44 acts are the only ground truth that exists for this task anywhere in the world, which is both why the US is the benchmark and why the benchmark is small.

The labels are act-level, but the codebooks work on chunks of documents, so the gold standard has to be projected from one grain to the other. That projection is itself a procedure with error in it, and the error turns out to matter more than any modelling choice in this section, so Section B documents how it was done and Section 5.1.1.4 quantifies what it cost.

This subsection proceeds in three parts. It first states what the benchmark covers and the one caveat that qualifies all of it. It then takes the two codebooks in pipeline order: measure identification (Section 5.1.1), then motivation and exogeneity (Section 5.1.2). Each runs the same sequence, behavioural tests, then zero-shot evaluation, then a manual error analysis that corrects for noise in the gold labels, and in each case the manual analysis is the gate rather than the automated metrics.

Figure 8 marks what the benchmark covers. The gap is as important as the coverage: it certifies the two LLM classification stages and says nothing about corpus construction or act identification, which are attested in other ways or not at all (Section 6).

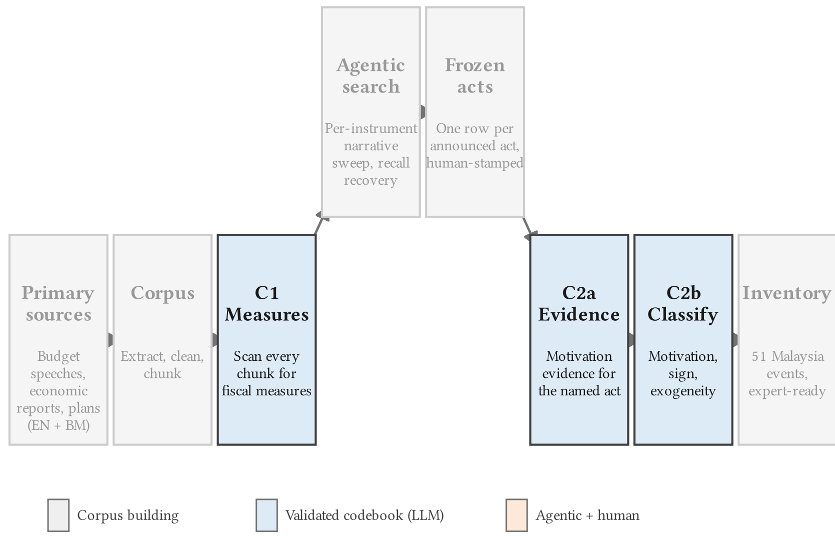

Figure 8: What the US benchmark covers. Stages carrying a formal validation record against US ground truth are lit; the rest are attested by other means.

One caveat frames everything that follows, and Section 2 has already named it. The published Romer–Romer shock list and the US source documents are almost certainly in the models' pretraining data, so strong US agreement is **consistent with memorisation** rather than proof against it (Carlini et al. 2022). This is why the behavioural battery includes an explicit memorisation probe, and why the burden of proof sits on the in-language Malaysia evidence in Section 5.2, where the documents are far less likely to have been seen during model training.

Both codebooks are put through the same battery of behavioural tests, and Table 12 serves as the reader's key to it. The column that matters is the last one: each test licenses a specific claim, and a codebook that fails one has lost the right to that claim regardless of how well it scores later. The tests divide into two groups by when they run. Tests I to IV run before any evaluation data is touched and ask whether the model can follow the codebook at all. Tests V to VII run alongside the error analysis and ask the harder question of whether it is following the codebook or something it already believed.

Table 12: The behavioural battery. What each test probes and what passing it licenses. Tests I–IV run before any evaluation data is touched; Tests V–VII run alongside the ablation and the manual error analysis (Halterman and Keith 2025).

Test	What it probes	Passing means
I. Legal outputs	Every response is one of the labels the codebook defines	The model can express itself in the codebook's vocabulary
II. Memorisation	Whether the model can recite the codebook's own definitions and instructions unprompted	Agreement later on is not recall of a document the model already knows
III. Example recovery	Whether the model reproduces the codebook's worked examples verbatim	The examples teach rather than leak
IV. Order invariance	Whether shuffling or reversing the class order changes the label	The label tracks the definition, not its position in the prompt
V. Exclusion criteria	Whether the codebook's stated exclusions are applied when the document or codebook is perturbed	Exclusion rules are enforced, not decorative
VI. Generic labels	Accuracy when informative label names are replaced by neutral ones	The model reads the definitions, not the label names
VII. Swapped labels	Whether the model follows the definition or the name when the two are put in conflict	The construct lives in the definition the researcher wrote

5.1.1 Measure identification

The measure-identification codebook decides whether a chunk substantively discusses a specific fiscal measure, and names the measures it finds. Table 13 gives the two labels it chooses between and the definitions it chooses by. Those definitions are the short form; each carries a set of inclusion and exclusion clarifications that do most of the discriminating work in practice, and the codebook is reproduced in full, exactly as the model reads it, in Section C. The same holds for every codebook in this paper: the body gives the labels and definitions, the appendix gives the artifact.

Table 13: Measure-identification classes. The two labels the codebook chooses between, and the definitions it chooses by.

Label	Definition
FISCAL_MEASURE	A chunk that contains, in any portion of its text, a substantive discussion of a specific legislative or executive action — whether enacted, proposed, or under consideration — that changes or would change fiscal liabilities or obligations.
NOT_FISCAL_MEASURE	A chunk in which no portion of the text substantively discusses a specific fiscal measure that changes or would change fiscal liabilities or obligations.

This codebook is tuned for recall, deliberately. It sits at the front of the pipeline, and a measure it misses is invisible to every stage after it, while a measure it wrongly surfaces is cheap: the act-identification pass discards it, and the motivation stage never sees it. So the operating point is set to maximise the fiscal policy available downstream, and precision is treated as recoverable rather than binding. The diagnostic benchmarks reflect that ordering — Tier 1 recall ≥95%, combined recall ≥90%, precision ≥70% — and none of them is a hard gate. The gate is the manual error analysis below.

5.1.1.1 Behavioural tests

Figure 9 shows the battery passing cleanly, and that is the expected result rather than an achievement: every frontier commercial model we tried clears these tests on this codebook. Their purpose is not to discriminate between models but to prove the harness is sound, that the schema, the parsing, and the token budget let the model express what the codebook asks for. They earned their keep anyway. An apparent Test I failure turned out to be an output-token cap truncating long measure lists rather than a codebook defect, and it was the test, not the evaluation, that isolated it.

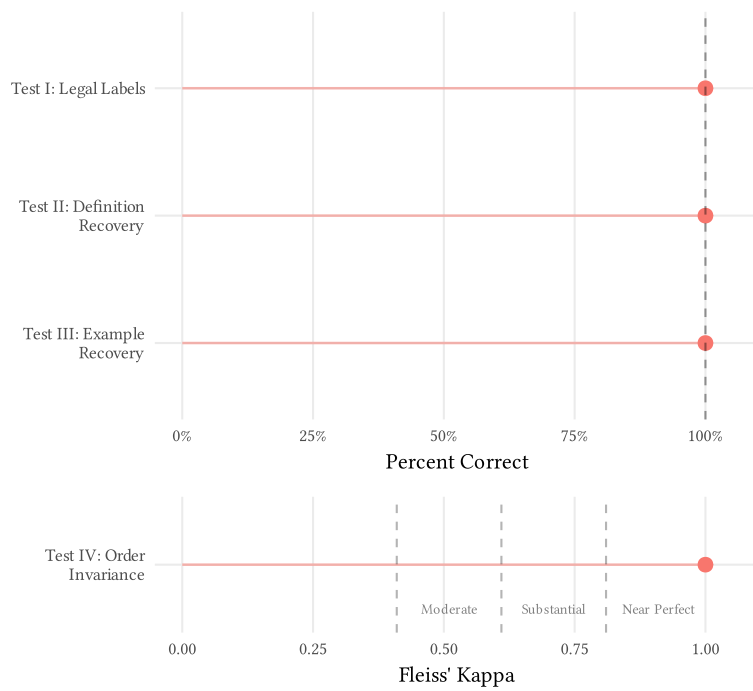

Figure 9: Measure identification, behavioural tests. Tests I–III as percent correct; Test IV as Fleiss' kappa across label orderings. See Table 12 for more information on the tests.

5.1.1.2 Zero-shot evaluation

Figure 10 represents the zero-shot evaluation against the projected gold labels, and it reports recall at two tiers, which needs a word of explanation. Because Romer and Romer's labels are attached to *acts* and the codebook reads *chunks*, positives have to be located in the corpus, and they are located two different ways with two different degrees of confidence. **Tier 1** chunks contain an act's narrative passage verbatim, so they are as close to a certain positive as this construction gets. **Tier 2** chunks merely *name* a known act, matched through whitespace-normalised substring search, decomposition of compound act names, and a small set of co-occurrence rules. Tier 2 is deliberately loose, because an act name appears in the corpus in many more places than its passage does, and Section B documents the construction in full.

The headline is that the codebook we tuned for recall is in fact stronger on precision: 94.7% precision against 80.0% combined recall, with Tier 1 recall at 88.2%. Both recall figures sit below their diagnostic benchmarks. Read at face value, that is a codebook that misses one positive chunk in five, which would be bad news given where it sits in the pipeline. The manual analysis below is what establishes that this is not what is happening.

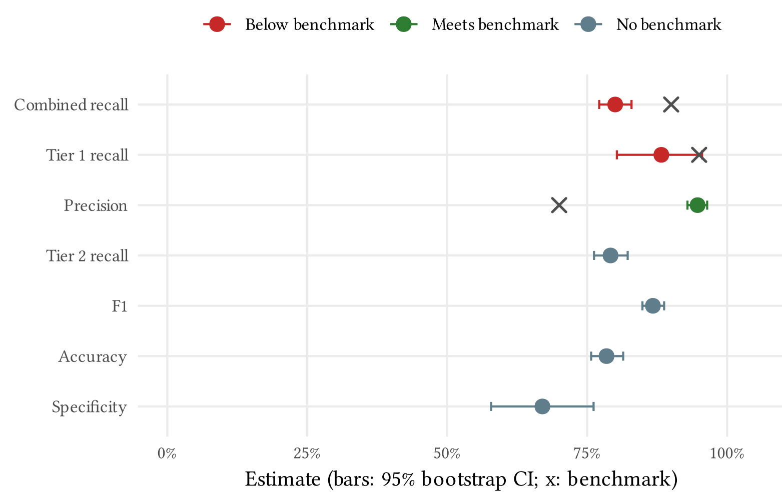

Figure 10: Measure identification, zero-shot evaluation. Point estimates with 95% bootstrap intervals; crosses mark the diagnostic benchmark where one is defined.

5.1.1.3 Codebook adherence

Accuracy alone cannot distinguish a model that read the codebook from one that already agreed with it. Figure 11 reports the three adherence tests that can: they perturb the codebook rather than the data, and ask whether the model's output moves with it. This is asking whether the model is right for the right reasons.

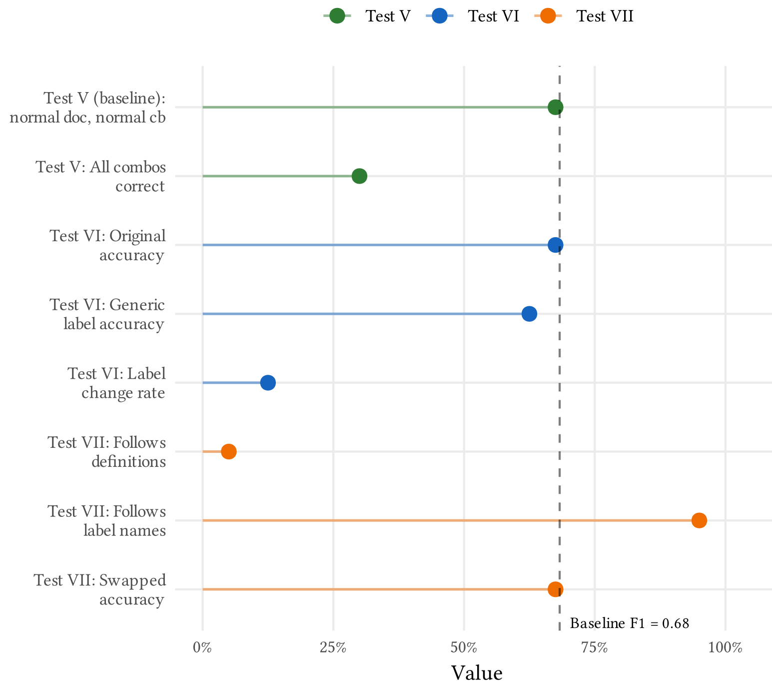

Figure 11: Measure identification, codebook-adherence tests. Tests V–VII against the full-codebook baseline. Test VI compares accuracy under informative and neutral label names; Test VII puts label names and definitions in conflict.

5.1.1.4 Manual error analysis and the label-noise correction

The gate is a human reading of every disagreement between model and gold label, sorted into the six-category framework of Halterman and Keith (2025): the model was right (A), the gold label was wrong (B), the document was corrupted (C), the model ignored the instructions (D), the model reasoned incorrectly (E), or something else (F).

Table 14: Manual error analysis. Distribution over the six-category framework for the measure-identification codebook.

Category	Count	Proportion
A: LLM correct	27	0.68
B: Incorrect gold standard	7	0.17
C: Document error	0	0.00
D: LLM non-compliance	0	0.00
E: Semantics/reasoning mistake	2	0.05
F: Other	4	0.10

The Category B count is the finding, not a footnote. Our US gold labels are not hand-coded chunk by chunk; they are attached by matching Romer–Romer act names against chunk text. Verbatim passage matches are reliable, but the second tier matches act-name *fragments*, and act-name fragments appear in book indexes, tables of contents, glossaries, and administrative appendices. Those chunks arrive carrying a positive gold label and no fiscal content whatsoever. When the model correctly declines to call a book index a fiscal measure, the evaluation scores it a false negative.

The label-generating process is noisier than the thing it is measuring, and that is a general hazard for LLM content analysis at this scale: automating gold label construction to make evaluation feasible imports exactly the errors the evaluation is meant to detect. It is the specific form, at scale, of the warning that runs through the economics text-as-data literature: measurement from text must be validated rather than assumed valid, and a validation is only ever as good as its own labels (Ash and Hansen 2023). The response is not to discard the noisy labels but to quantify them. The manual analysis identifies which disagreements are label noise, and the gate is recomputed with those chunks excluded from the denominator, as Table 15 and Table 16 report. Ambiguous cases are kept in the denominator rather than excluded, so the correction is conservative.

Table 15: Measure identification, corrected confusion matrix. Counts after excluding the chunks the manual analysis judged mislabelled in the gold standard.

	Actual positive	Actual negative	Total
Predicted positive	14	4	18
Predicted negative	2	13	15
Total	16	17	33

Table 16 reports the corrected rates against the diagnostic benchmarks. The correction moves the picture substantially: on the 33 chunks that survive it, recall rises to 87.5% and Tier 1 recall to 80.0%, against raw figures of 80.0% and 88.2%. Neither reaches its benchmark. We nonetheless took the codebook forward, and the reasoning should be visible rather than buried in a "passed". The benchmarks were set as diagnostics, not gates, precisely because we anticipated that the projected gold labels would be noisy, and the manual analysis confirmed that anticipation in a specific and correctable form. The effective sample after correction is small enough that these rates carry wide intervals. And the quantity the pipeline actually depends on is whether real fiscal content reaches the identification stage, which the Malaysia deployment then tests directly: the agentic pass found the corpus richly populated, and the misses it did record were rate-path steps rather than whole measures. That is a judgement call, made once, recorded here, and open to disagreement.

Table 16: Measure identification, corrected performance. Label-noise-adjusted rates against the diagnostic benchmarks. Recall is the binding quantity here; precision is recoverable downstream.

Metric	Estimate	Target	Status
Accuracy	81.8%	—	—
Precision	77.8%	≥ 70%	Pass
Recall	87.5%	≥ 90%	Below gate
Tier 1 recall	80.0%	≥ 95%	Below gate
Tier 2 recall	100.0%	—	—
Specificity	76.5%	—	—

5.1.2 Motivation and exogeneity

5.1.2.1 Why this is two codebooks

Assigning a Romer–Romer motivation label to a passage is two tasks wearing one name. The first is evidentiary: find, in a document that discusses many things, the sentences stating why *this* measure was enacted. The second is interpretive: decide which of the four motivation categories that stated reason belongs to, and whether it makes the act exogenous to current output.

Fusing them into one prompt makes failures uninterpretable — a wrong label could be a reading failure or a reasoning failure, and the transcript cannot tell you which. So the stage is split. An **extraction** codebook returns the motivation evidence for a named act as verbatim quotes. A **classification** codebook sees only those quotes and returns the label, the sign, and the exogeneity call. The split buys three things: errors localise to a stage, the extracted evidence becomes the audit trail a reviewer reads instead of the label, and the classification step can be evaluated on clean evidence independently of retrieval quality.

The split creates a problem for validation, though, and the way we resolve it is worth stating because it is the reason Section 4.5 happened. The Halterman and Keith framework is built to evaluate **classification**: its behavioural tests perturb label names and definitions, its zero-shot stage compares a predicted label against a gold label, and its error taxonomy sorts disagreements between two labels. None of that has a natural analogue for a free-form extraction step whose output is a set of quotes. There is no gold set of correct quotes to score against, and no meaningful confusion matrix over spans of text.

So the framework is applied where it fits, to the classifier, and the extraction step is evaluated only through its effect on the classifier's output. In practice that means the gate is applied to the composition: extraction runs on real chunks, and the classifier is scored on whatever evidence extraction actually produced, errors included, rather than on clean evidence hand-picked for it. The extraction codebook carries its own behavioural and zero-shot record but no separate error analysis, because the only errors we can adjudicate are the ones that reach a label.

That choice has a cost, and Section 4.5 is what it bought. The defect lived in the *interface* between the two stages, not inside either one, and no metric either stage carries could have surfaced it. Scoring the composition means such a defect shows up as a slightly worse number rather than as a diagnosis, and here it did not show up at all, because the composition was scored on US chunks where the defect is inert.

Table 17 gives the four motivation labels and their definitions as implemented. The order matters: the first two are responses to the economy and therefore endogenous, and the last two are the pair the shock series keeps.

Table 17: Motivation classes. The Romer–Romer four-way taxonomy as implemented. The first two are endogenous to current output; the last two are exogenous and are the ones that survive into the shock series.

Label	Definition
SPENDING_DRIVEN	A fiscal action motivated by a contemporaneous change in government spending — a revenue measure taken to pay for, or to offset the expansionary effects of, an associated spending increase.
COUNTERCYCLICAL	A fiscal action designed to return output growth to normal — taken in response to current or prospective economic conditions that are expected to push growth below (or above) its normal rate in the near future.
DEFICIT_DRIVEN	A fiscal action — typically a revenue increase, sometimes coupled with spending cuts — designed to reduce an inherited budget deficit resulting from past economic conditions and past policy decisions, rather than from any contemporaneous spending change or current cyclical conditions.
LONG_RUN	A fiscal action aimed at raising long-run growth or pursuing other structural goals — improved incentives, greater efficiency, fairness, or a philosophical preference for smaller government — rather than at responding to current or prospective short-run economic conditions.

5.1.2.2 Behavioural tests and zero-shot evaluation

The behavioural battery is the same one Table 12 describes, and Figure 12 shows the classification codebook clearing it. Test II carries more weight here than it did for measure identification, because it is the memorisation probe and this is the codebook whose ground truth is a published, widely-cited act list.

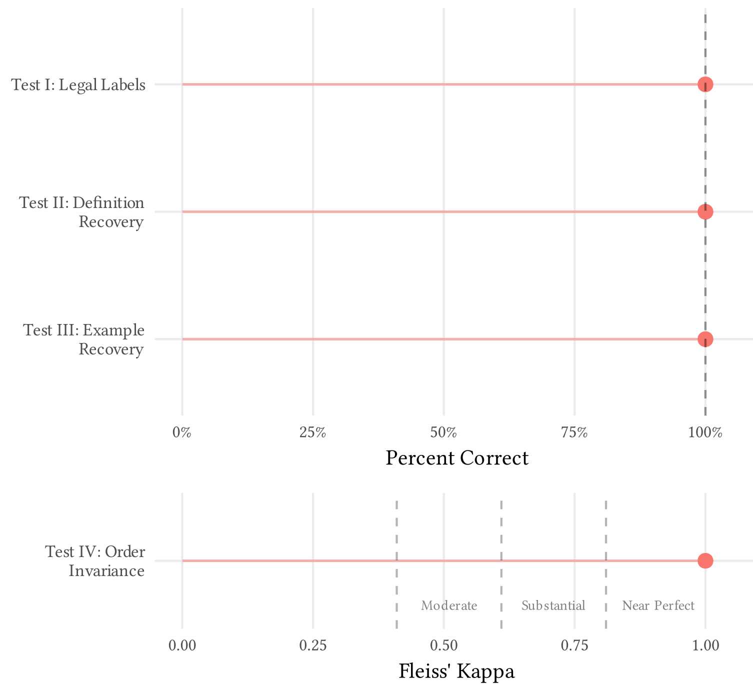

Figure 12: Motivation classification, behavioural tests. Tests I–III as percent correct; Test IV as Fleiss' kappa across label orderings. Test II is the memorisation probe for the classification codebook.

Figure 13 shows the zero-shot evaluation. Two of these metrics are the ones a downstream user depends on. Exogenous precision, the share of acts handed over as exogenous that really are, lands at 80.0% against an 85% gate, though on a sample small enough that the interval runs from 63% to 96%. Sign accuracy on truly-exogenous acts, at 91.3%, clears its 90% gate. The rest of the panel is diagnostic: the four-way task is harder than the binary one, and exogenous recall in particular is weak, which matters much less than precision because a missed exogenous act shrinks the series while a wrongly-included one contaminates it.

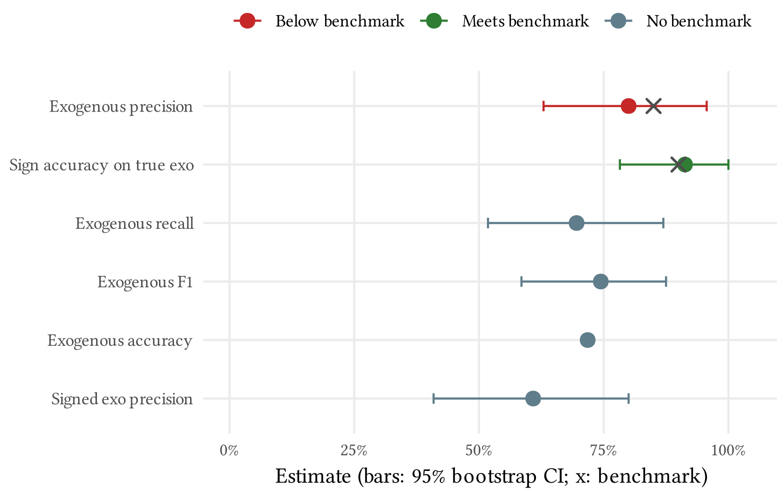

Figure 13: Motivation classification, zero-shot evaluation. Point estimates with 95% bootstrap intervals; crosses mark the gate where one is defined. Exogenous precision and sign accuracy on true-exogenous acts are the two gates that matter downstream.

Figure 14 reports the adherence tests, and it is the most consequential exhibit in this section. It asks not how often the classifier is right but whether it is reading our codebook to get there, by perturbing the codebook and watching what happens to the output.

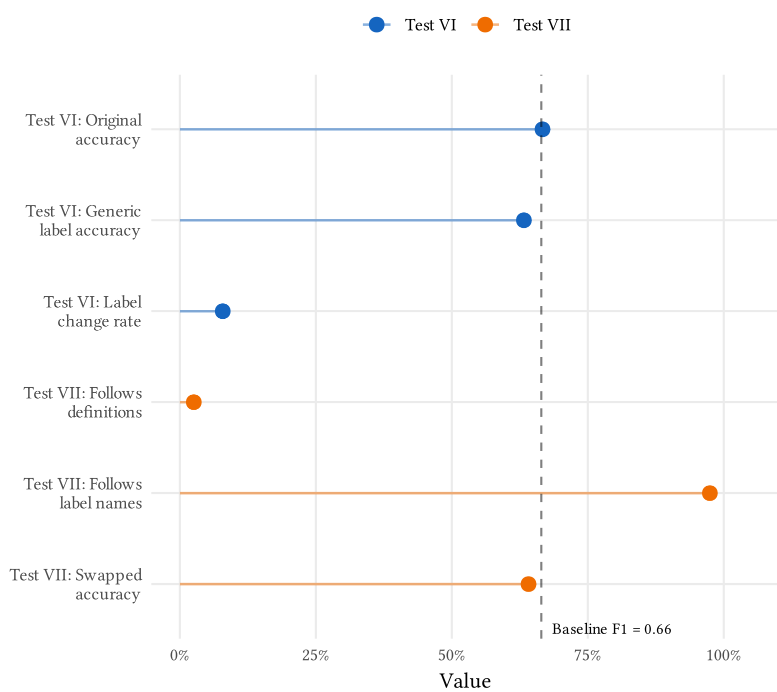

Figure 14: Motivation classification, codebook-adherence tests. Tests V–VII against the full-codebook baseline. Test VI compares accuracy under informative and neutral label names; Test VII puts label names and definitions in conflict; the ablation strips codebook components one at a time.

The result is uncomfortable and we report it rather than bury it. When label names and label definitions are put in conflict (Test VII), the model follows the *names* almost always and the definitions almost never. Read together with a small ablation penalty — stripping the codebook's definitions and clarifications costs only a few points of accuracy, and even the fully stripped condition retains most of the four-way task — the natural reading is that the model arrives already knowing what "countercyclical" and "long-run" mean in fiscal policy, from pretraining, and that our codebook is calibrating a prior rather than teaching a construct.

That is good news for transfer and bad news for control. It is good news because a prior learned from the general fiscal-policy literature is not tied to US institutions, which is the property cross-country deployment needs. It is bad news because a researcher who redefines a category away from its conventional meaning should not assume the model will follow, and because agreement on US data is weaker evidence than it appears. It is the sharpest reason the Malaysia evidence, not the US benchmark, has to carry the argument.

There is a name for the property this result is about. Asirvatham et al. (2026) separate two requirements an LLM measurement has to meet: **accuracy**, agreement with a benchmark, and **directness**, the property that the rating comes from the text being rated rather than from something merely correlated with it. Test VII is a directness result, and it is the test we do worst on. Their diagnostic is signal-stripping: remove the rated content and confirm the rating collapses. The analogue here is to mask the stated motivation in a chunk and re-classify, and if the exogeneity call survives, the model is reading the act's name, its era, and its sponsor rather than its rationale. We have not run that test. It is the single most informative check missing from this paper, and Table 22 records it as such.

The finding also puts the encouraging annotation literature in context. Where Gilardi et al. (2023) and Törnberg (2025) find models matching or beating human coders, the constructs at issue are conventional ones a model has seen defined many times over, and Ziems et al. (2024) find the advantage narrows precisely where labels are bespoke and taxonomic. A pretrained prior is the most plausible reason our classifier does well on Romer and Romer's categories, which are canonical in fiscal economics. It is equally a reason to expect less of any category a researcher invents.

5.1.2.3 Manual error analysis and the label-noise correction

As with measure identification, the gate is a human reading every disagreement between the model and the gold label and sorting it into the same six categories. Table 18 gives the distribution, adjudicated on the composed extraction-and-classification chain rather than on either codebook alone.

Table 18: Manual error analysis. Distribution over the six-category framework for the motivation codebooks, adjudicated on the composed extraction-and-classification chain.

Category	Count	Proportion
A: LLM correct	24	0.62
B: Incorrect gold standard	2	0.05
C: Document error	0	0.00
D: LLM non-compliance	0	0.00
E: Semantics/reasoning mistake	2	0.05
F: Other	11	0.28

The label noise here has a different source than in Section 5.1.1.4 and is harder to correct. Motivation gold labels come from Romer and Romer's own classifications, which are act-level judgements published with their reasoning. Where a chunk discusses an act only glancingly, or where the act's motivation is genuinely mixed, the act-level label is not a defensible chunk-level target — and unlike an index page, there is no mechanical signature that marks such a case. Every Category B act here is of that kind: the act carries two motivations in its own stated reasons, and the gold label follows a tie-breaking convention rather than the codebook. Adjudicating those is itself an interpretive act, so the Category B count here is less mechanical and more contestable than the measure-side one. We again exclude only what the manual pass judged mislabelled and keep the ambiguous cases in the denominator.

Table 19: Motivation classification, corrected confusion matrix. Exogenous-versus-endogenous counts after excluding the acts the manual analysis judged mislabelled or undecidable in the gold standard.

	Actual positive	Actual negative	Total
Predicted positive	15	3	18
Predicted negative	7	12	19
Total	22	15	37

Table 20: Motivation classification, corrected performance. Label-noise-adjusted rates against the two gates that bind: exogenous precision, and sign accuracy on acts that are truly exogenous.

Metric	Estimate	Target	Status
Exogenous precision	83.3%	≥ 85%	Below gate
Exogenous recall	68.2%	—	—
Sign accuracy on true-exogenous	95.5%	≥ 90%	Pass
Joint label-and-sign accuracy	64.9%	—	—

Exogenous precision is the quantity a downstream user depends on: it is the share of the acts we hand over as exogenous that really are. It lands just below its gate, on an effective sample small enough that the interval comfortably contains the target — a result we report as borderline rather than as a pass. Sign accuracy on truly-exogenous acts clears its gate. These two corrected numbers, and the correction procedure that produces them, are the central methodological output of the US benchmark.

5.2 Malaysia

Do the same codebooks, frozen, still work in a country with no labelled data?

This is the question the project exists to answer, and it cannot be answered the way the last section was. There is no Malaysian equivalent of the Romer–Romer act list, so there is nothing to compute an accuracy against. That is not a temporary inconvenience: if a validated narrative series already existed for a country, the pipeline would have little to offer it. The absence of ground truth is the condition the method has to work under, not an obstacle on the way to working under better conditions.

What replaces an accuracy rate is four separate lines of evidence, none sufficient alone. The first is an **internal consistency test** that needs no ground truth at all, exploiting a property of the corpus to check that the pipeline reasons about fiscal substance rather than about English. The second is a set of **worked disagreement cases**, printed in full so a reader can judge the pipeline's reasoning directly rather than through a summary statistic. The third is the **Vegh and Vuletin (2015) statutory-rate panel**, which is an external referent for one narrow part of the output and no part of the motivation. The fourth is **expert agreement**, which is the only external check on the judgement itself and is currently being collected.

The third deserves a word, because it qualifies the claim just made. No external referent exists for *motivation*, which is the quantity the method is about. One does exist for the *rate path*, which is a by-product the method has to get right along the way, and it turns out to be checkable against published data for two of the five instruments. That is a narrower thing than validation of the pipeline, and it is more than nothing.

A fourth line exists in principle and we cannot yet use it. Aruoba and Drechsel (n.d.), identifying monetary policy shocks from narrative records, validate not against an existing shock list but against forecast-error predictability and against whether the recovered series produces theory-consistent impulse responses. Neither test needs labels, which makes the strategy the natural one for a country like Malaysia. Both need magnitude and quarterly timing, which Table 22 records as unbuilt. So the strategy is available to this method and not yet to this dataset, and it is a further reason Section 6 treats magnitude as a binding gap rather than a refinement.

5.2.1 Cross-language (EN/BM) consistency

The test exploits an accident of the corpus. Ten Economic Reports were published in both English and Bahasa Malaysia: the same fiscal year, the same underlying acts, two languages. A pipeline that reasons about fiscal substance should return comparable inventories from both. A pipeline that has learned English surface patterns, or that is retrieving remembered facts cued by English phrasing, should not.

This is a sharper instrument here than it would be in most applications, for two reasons. Multilingual evaluations consistently find degradation on lower-resource languages relative to English (Ahuja et al. 2023; Rathje et al. 2024), so equivalence is a real hypothesis and not a foregone conclusion. And the paired-document design isolates the language effect from item difficulty, since both versions describe identical events, which is exactly the design the multilingual benchmarks adopt to separate the two (Singh et al. 2024; Xuan et al. 2025).

Because no ground truth exists on either side, the comparison is distributional: measure counts, act counts, and the motivation label mix, rather than a matched act-by-act agreement rate. That is a weaker test than matched agreement would be, and we prefer it to the alternative, which would be to build the matching with a third model and then validate two unvalidated things against each other. Comparable distributions are consistent with language-robust reasoning without proving it; sharply different ones would have been decisive evidence against.

Figure 15 takes the first stage of the pipeline: how many distinct fiscal measures the codebook surfaces from each report year, read in each language. The question it answers is whether the scan sees the same amount of fiscal content in a document regardless of the language it is written in.

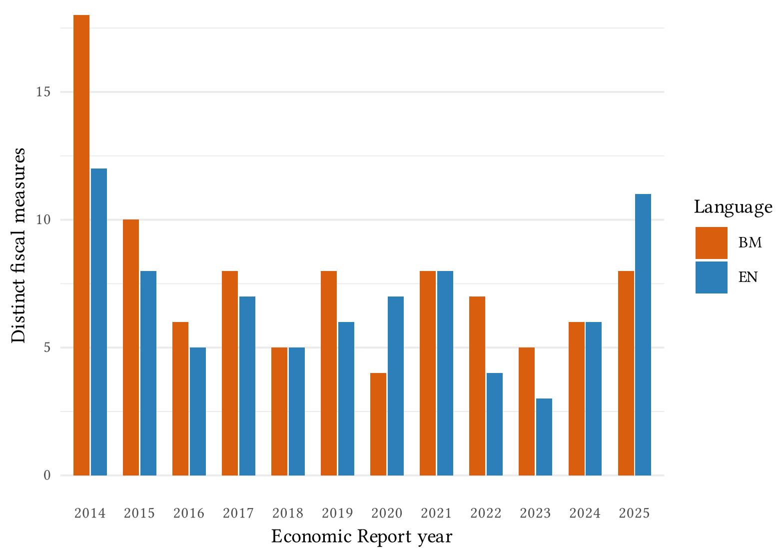

Figure 15: Measure identification across languages. Distinct fiscal measures surfaced per Economic Report year, by language. Comparable counts indicate the extraction step is language-robust.

Figure 16 carries the test through to the end of the chain, comparing the act inventories the two language runs produce and the motivation labels attached to them. This is the more demanding comparison, because it accumulates whatever divergence the earlier stages introduced.

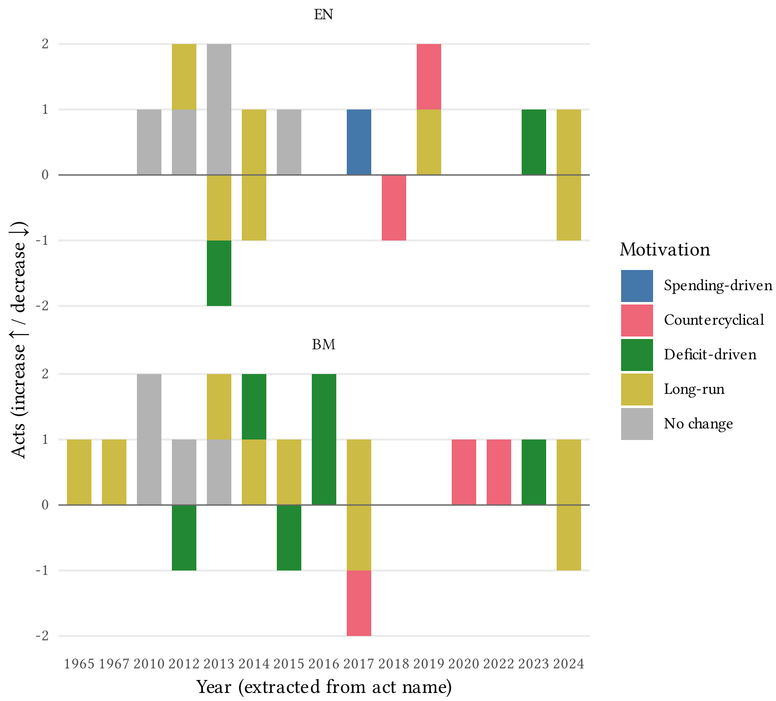

Figure 16: Act inventory across languages. Acts by year and motivation, faceted by language. Comparable structure across English and Bahasa Malaysia is the signal that the codebooks reason consistently across languages.

5.2.2 Worked acts (disagreement cases)

Aggregate evidence has a limit: it can show that two runs agree without showing whether either is reasoning well. So we do what Romer and Romer (2010) do in their own Exhibits 1 and 2, and print acts in full for the reader to judge.

The three below are chosen adversarially. Each is a case where the pipeline's two independent reads, the preliminary narrative read formed during act identification and the classification codebook's verdict, **disagree** about exogeneity. These are not the pipeline at its best; they are the cases where it cannot settle the question itself, and they are the ones an expert has to adjudicate. Printing them is the point, because a method that hides its disagreements is asking to be trusted rather than checked.

Read them with Section 4.5 in mind. Two of the four labels that section identifies as built on the wrong act's evidence are corporate-tax rows, and the exhibits below are drawn from the June freeze, so a disagreement shown here may be an artifact of that defect rather than a genuine interpretive tension. That is itself instructive: without the evidence chain printed alongside the label, a reader would have no way to tell the two apart.

Exhibit 2 — Corporate income tax rate reduction 30% to 28% (Budget 1998)

Shock ID: MY-CIT-03 | **Tax:** CIT | **Announced:** 1997 | **Effective:** 1998
Rate change: 30% ⟶ 28% (-2 pp; Cut)
Classification (C2b): Endogenous — Countercyclical
Signed effect on liabilities: NA *(no single sign; raw model output: "mixed")*

Identification. 30->28 reduction announced in the 1998 Budget (Oct 1997), effective YA1998. Evidence: BNM Annual Report 1997.

Motivation (C2b). The evidence consistently shows that the 1998 Budget and accompanying fiscal measures were designed to restore economic stability and return growth to normal in response to the 1997 Asian financial crisis and regional currency turmoil. Policymakers explicitly stated the goal was to 'restore stability to overcome disruptions', 'address imbalances in the economy', and 'restore macroeconomic stability and investor confidence' following the July 1997 currency crisis. While the package included some long-run structural elements (tax incentives for productivity and competitiveness), the predominant and repeatedly stated motivation across all evidence was cyclical stabilization in response to the regional financial shock and its effects on Malaysia's economy (currency depreciation, banking vulnerabilities, current account deficit, inflation). The fiscal measures were explicitly framed as responses to current economic conditions and external shocks, not as structural reforms or deficit reduction from inherited imbalances. This satisfies the countercyclical criterion: actions taken to offset developments that would cause output growth to differ from normal.

> reduce the cost of doing business and strengthen competitiveness ... reduced to 28% in 1998 compared with 30% each for Indonesia and Thailand

Preliminary narrative read: exogenous — **differs from the C2b verdict; this is exactly the kind of call an expert must adjudicate.**

The narrative pass read this rate cut as exogenous; the classifier read the surrounding crisis framing and called it countercyclical. Both readings are defensible from the corpus, which is what makes it an adjudication case.

Sources: Bank Negara Malaysia Annual Report (1997, en)

Exhibit 3 — Fuel-subsidy restructuring (June 2008)

Shock ID: MY-SPEND-04 | **Category:** Subsidies | **Announced:** 2008 | **Effective:** 2008Q2
Change: Decrease — Retail fuel prices raised ~40.4% in June 2008 as the global crude-oil spike made the subsidy fiscally unsustainable; partial offsets (rice subsidy, fishermen/transport fuel subsidy, motorist fuel cash rebate).
Classification (C2b): Exogenous — Deficit-driven
Signed effect (C2b): decrease (−)

Identification. Distinct from the 2010 structural rationalisation (reviewer decision: separate row). Driven by the 2008 oil-price spike / financing pressure on the subsidy bill -> cyclical/ price-driven -> endogenous.

Motivation (C2b). The fuel subsidy restructuring was motivated primarily by fiscal sustainability concerns, not by contemporaneous spending changes or cyclical conditions. Evidence explicitly states the government restructured subsidies to 'avoid prolonged high fiscal deficit' and because subsidy costs had risen unsustainably from RM1.7 billion (2002) to RM7.5 billion (2007), projected to reach RM28 billion in 2008, 'threatening fiscal sustainability.' While mitigation measures (cash rebates, targeted subsidies) were introduced to cushion the impact on citizens, these were secondary responses to the primary deficit-reduction motivation. The action addresses an inherited fiscal burden from past subsidy commitments at volatile global oil prices, not a contemporaneous spending increase or cyclical downturn.

> The fuel subsidy restructuring exercise undertaken by the Government in June caused fuel prices to increase by 40.4%, resulting in a sharp increase in the headline inflation rate, which peaked at 8.5% in July.

Preliminary narrative read (Das screen): endogenous — **differs from the C2b verdict; this is exactly the kind of call an expert must adjudicate.**

The narrative pass applied the two-condition spending screen and read the subsidy restructuring as endogenous; the classifier read it as deficit-driven, hence exogenous. The disagreement is about whether fiscal-pressure motives count as cyclical.

Sources: Bank Negara Malaysia Annual Report (2008, en); Economic Report / Tinjauan Ekonomi (2009, en)

Exhibit 4 — Companies-relocating-to-Malaysia incentive (PENJANA / Budget 2021)

Shock ID: MY-INCENT-12 | **Base:** CIT | **Category:** Preferential rate | **Announced:** 2020 | **Effective:** 2020
Change: Non-rate incentive (Cut) — Relocation incentive introduced under PENJANA (Jun 2020) / Budget 2021: 0% tax for new manufacturing companies (10-15 yr) and 10% for existing companies (5 yr), or 100% Investment Tax Allowance
Classification (C2b): Endogenous — Countercyclical
Signed effect (C2b): decrease (−)

Identification. Seed PREFERENTIAL_RATE regime. Relocation incentive introduced under PENJANA/Budget 2021. Per reviewer decision the 2020 introduction and the 2023 extension are SPLIT into two rows (this row = introduction).

Motivation (C2b). The act's primary stated motivation is to 'increase disposable income' and 'assist tax payers affected by the current economic situation' (2020 COVID-19 downturn) and to 'spur economic recovery' through investment incentives and multiplier effects. These are direct statements of intent to offset current cyclical weakness and return growth to normal, not to raise long-run potential output or pursue structural reform. The multiple references to alleviating costs and encouraging investment in response to the economic crisis, combined with the 2020 timing during Malaysia's COVID-19 recession, establish countercyclical motivation as the predominant rationale.

> New and existing companies that relocate their business or manufacturing activities from abroad to Malaysia are eligible for the following tax incentives [Review of Tax Incentives for Companies Relocating Their Operations to Malaysia and Undertaking New Investments].

Preliminary narrative read: exogenous — **differs from the C2b verdict; this is exactly the kind of call an expert must adjudicate.**

An investment incentive announced inside a pandemic recovery package. The narrative pass weighted the long-run investment-promotion objective, the classifier weighted the crisis vehicle it arrived in.

Sources: Budget Speech / Ucapan Bajet (2021, en)

5.2.3 The Végh–Vuletin statutory-rate panel

This section opened by saying no external referent exists for Malaysia. For motivation that is true and will stay true. For the *rate path* it is not: the statutory top rates the narrative pass reconstructs, shown as Figure 3 , also appear in the hand-built cross-country rate panel Vegh and Vuletin (2015) assemble, which covers Malaysia over this period. That comparison is the most direct external validation the magnitude side admits. It is run here.

The benchmark is well chosen for more than convenience. It is the same series whose procyclicality finding Section 7 takes as its starting point, so the check and the use case are run against one another's data: we validate our rate paths against the panel, then show what the panel's own cyclicality statistic cannot resolve on the country it covers.

The check is narrow and worth stating precisely. It scores whether we recovered the right rate, in the right year, on the two instruments that have a headline rate. It says nothing about motivation, about spending, or about incentives, because a rate panel carries a number per year and no record of why it moved.

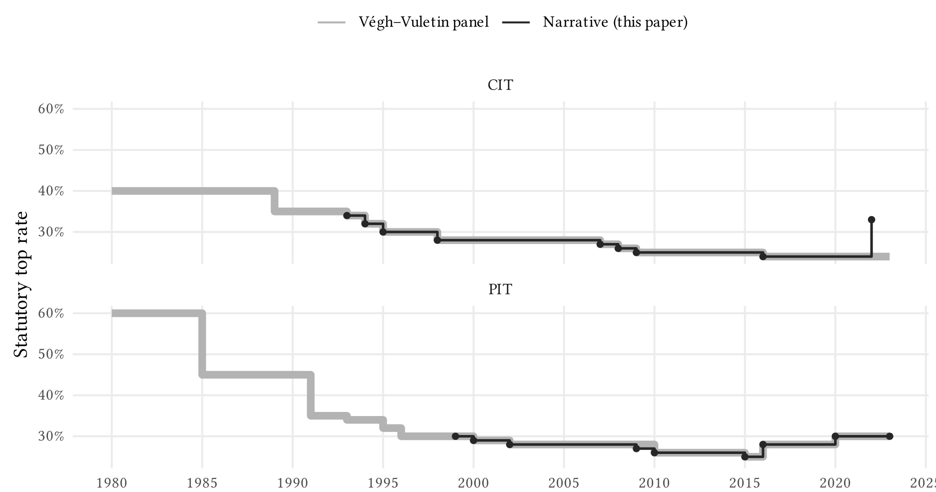

Figure 17: Narrative rate paths against the Végh–Vuletin panel. The thick grey line is their published series (Vegh and Vuletin 2015); the thin dark line with markers is the path implied by the acts this pipeline identified. Exact agreement reads as the dark line sitting inside the grey one. The dark line starts where the corpus becomes dense, not where the rate series starts.

Table 21 scores it event by event. Three patterns, and each says something different.

Where the corpus is dense, the paths agree exactly. Every corporate rate change the panel records from 1994 onward is matched by an act we identified, at the same year and with the same change in percentage points. The personal rate is comparable from 1999. This is a stronger result than a correlation would be, because the match is on the level and the year, not on co-movement.

Every miss predates the corpus. The unmatched panel changes are all early: the corpus holds a single document in each of several years before the mid-1990s and only becomes dense afterwards, as Figure 7 shows. That is a statement about what we gave the pipeline to read, not about how it read. It is also the cleanest available illustration of the point Section 4.3 makes about recall: the ceiling is set by acquisition.

Every extra is something the panel cannot represent. The acts we identify that the panel does not are not false positives. A one-off corporate surcharge, a change to the imputation system, a cut to a middle band, a restructure that leaves the top rate untouched: each moves real liabilities and none of them moves a headline top rate, so a rate panel is blind to them by construction. This is the asymmetry that motivates the whole exercise. The rate panel is the better instrument for the narrow question of what the top rate was; the narrative inventory is the only one of the two that can see the rest of the tax policy.

Table 21: Event-level concordance. Every rate change recorded by either source; the two numeric columns are the change in percentage points. Matched rows agree on year and magnitude. The timing row is one panel step the narrative pass resolves into two annual steps, agreeing on the level by the end of it.

Tax	Year	Outcome	Panel	Ours	Note
CIT	1989	External only	−5	—	Predates corpus coverage
CIT	1993	External only	−1	—	Predates corpus coverage
CIT	1994	Matched	−2	−2	
CIT	1995	Matched	−2	−2	
CIT	1998	Matched	−2	−2	
CIT	2007	Matched	−1	−1	
CIT	2008	Matched	−1	−1	
CIT	2008	Narrative only	—	—	No change to the headline top rate
CIT	2009	Matched	−1	−1	
CIT	2016	Matched	−1	−1	
CIT	2022	Narrative only	—	+9	Moves liabilities without moving the headline top rate
PIT	1985	External only	−15	—	Predates corpus coverage
PIT	1991	External only	−10	—	Predates corpus coverage
PIT	1993	External only	−1	—	Predates corpus coverage
PIT	1995	External only	−2	—	Predates corpus coverage
PIT	1996	External only	−2	—	Predates corpus coverage
PIT	2000	Matched	−1	−1	
PIT	2002	Matched	−1	−1	
PIT	2010	Timing or aggregation	−2	−2	
PIT	2015	Matched	−1	−1	
PIT	2016	Matched	+3	+3	
PIT	2018	Narrative only	—	—	No change to the headline top rate
PIT	2020	Matched	+2	+2	
PIT	2023	Narrative only	—	0	No change to the headline top rate

Two limits on how far this evidence reaches. It covers corporate and personal income tax only: the panel carries no consumption-tax rate for Malaysia, so the GST introduction and repeal, which are the largest tax events in the inventory, get no external check at all. And a matching rate path is consistent with, but does not establish, a correct reading of the document, since a model could in principle recover a well-known rate history without reading anything. The cross-language test above is the evidence that bears on that, and it is separate.

5.2.4 The expert-agreement protocol

The external check is expert review, and its design has to answer a question that does not arise when ground truth exists: what does agreement with a human *mean*, when the human is not an oracle either?

The protocol asks regional fiscal-policy experts to code sampled acts independently, using the same codebook definitions the model was given, and then compares the two codings. Two headline targets are pre-registered: at least 80% agreement on measure identification, and at least 70% on motivation. Those thresholds are lower than an accuracy target would be, deliberately, because they are **parity** targets rather than correctness targets. Experts disagree with each other about motivation, sometimes substantially, and a model that agrees with one expert as often as a second expert would is doing as well as the task allows. Following Asirvatham et al. (2026), the right comparison is therefore model-versus-consensus against leave-one-out human-versus-consensus, with expert-versus-expert agreement reported as the ceiling. Without that ceiling, a 70% figure is uninterpretable: it could be a weak model or a genuinely contested construct, and only the human baseline separates them.

Three further features of the design are worth stating. Review is **risk-stratified** rather than uniform: acts predicted exogenous in a recession year are reviewed in full, because that is the cell where an error does the most damage to any downstream estimate, as are acts carrying either of the two labels the error analysis found most confusable. The remainder is spot-checked. The protocol also asks a question the dataset cannot ask itself, namely which significant acts are **missing**, which is the recall assessment Section 4.3 defers to. And it separates genuine model error from irreducible ambiguity: where two experts disagree with each other, the case is recorded as ambiguous rather than scored against the model.

The first of those strata is more load-bearing than it looks. Acts predicted exogenous in a downturn are precisely the cell the decomposition in Section 7 rests on, and Section 7.4 reports how few of them would have to be relabelled to overturn it. So the review burden that section implies is not open-ended: it is a specific, countable set of acts, and the protocol was already routing them to full review before anyone had computed the number.

One gap in the protocol as currently written should be closed before it is run, because Section 7 depends on it. The protocol selects acts for review by purposive triage — every act predicted exogenous in a recession year, every act carrying one of the two confusable motivation labels, and a convenience spot-check of the rest. That is the right design for *finding* errors and the wrong design for *correcting* an estimate: debiasing a downstream regression against a validation set requires that set to be a probability sample, with known inclusion probabilities (Angelopoulos et al. 2023; Egami et al. 2023). The two purposes need either two samples or one design that serves both.

6 What is validated, and what is not

The pipeline's stages rest on very different evidence, and the difference is easy to lose once the output is a single tidy table. Figure 18 inverts Figure 8 : it lights the stages the US benchmark does *not* cover, and Table 22 states what each of them rests on instead.

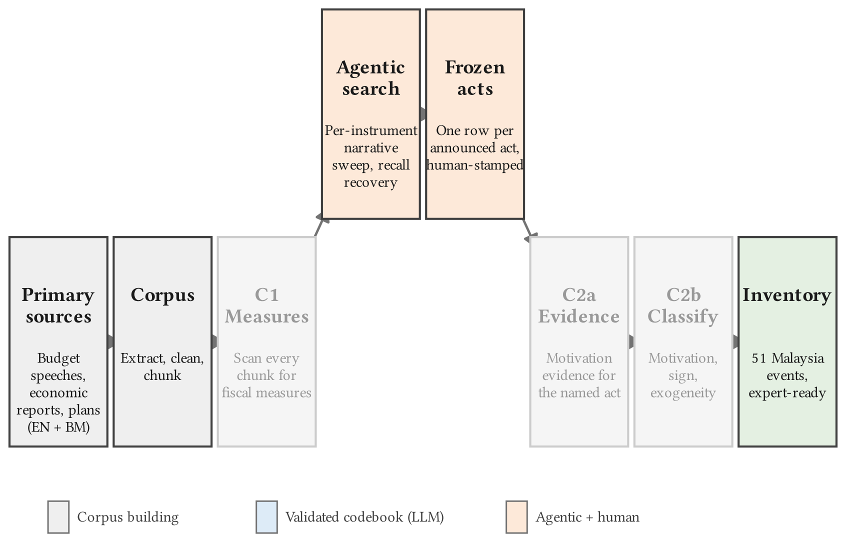

Figure 18: What the US benchmark does not cover. The complement of Figure 8 . These stages are attested by construction, by audit, or not yet at all.

Table 22: Validation coverage. What each stage of the pipeline rests on, and what is still missing. Nothing in the deliverable's exogeneity column is an adjudicated finding.

Stage	Evidence we have	Evidence we lack
Corpus construction	Extraction QA; page and chunk counts reproducible from the pipeline	No recall audit over the universe of documents that exist
Measure identification	Full behavioural, zero-shot and error-analysis record on 44 US acts	Deployment-corpus performance; the top-ranked-measure filter discards 84.5% of mentions
Act identification	Per-instrument recall scorecards, frozen; human stamp on every event	No gold standard, so no recall or precision rate — only an audit of what was searched
Act consolidation	Automated clustering built and scored against 49 US reference acts	Rejected for deployment, not validated for it
Motivation and sign	Full record on US acts; label-noise-corrected gates	Validated on single-act chunks, deployed on multi-measure speeches; no signal-stripping test of directness
Magnitude	Traceable to a quoted rate or amount; CIT and PIT top-rate paths reproduce the Végh and Vuletin (2015) statutory-rate panel exactly over the corpus window	No codebook, no behavioural tests; the external check reaches only headline top rates, not surcharges, band changes, consumption taxes or spending amounts
Timing	Announced and effective year, from the document; effective years match the Végh and Vuletin panel where both cover	No quarterly timing; the annual grain limits the downstream use case
Exogeneity of the output	Two independent preliminary reads per act, and their disagreements flagged	Expert adjudication not yet run; the column is a model read

Three entries deserve emphasis because they are the ones most likely to be over-read.

The exogeneity column is an input, not a finding. Every act carries two preliminary reads — one from the narrative identification pass, one from the validated classification codebook — and where they disagree the act is flagged rather than resolved. Those disagreements are the worked exhibits in Section 5.2. Adjudicating them is the expert's job and it has not happened yet.

Magnitude has a partial record and timing has none. Both are extracted agentically alongside each act, and the codebooks that would validate them are not built. What magnitude now has is a scope-limited external check: Section 5.2.3 shows the corporate and personal top-rate paths reproducing the Vegh and Vuletin (2015) statutory-rate panel exactly wherever the corpus is dense. That covers the rate-bearing acts on two instruments. It does not cover the consumption taxes, the surcharges and band changes a top-rate series cannot represent, or any spending amount, and those are most of the magnitude column. Quarterly timing has no check of any kind.

The deployment distribution was never validated. This is the general form of Section 4.5 and the most transferable lesson here. The codebooks were validated on US chunks that discuss one act each and deployed on Malaysian budget speeches that discuss many. That gap produced a real defect that no US test could have caught, and it is a gap any cross-country transfer of this kind will have in some form.

The same accounting is worth doing for the closest related work, because the comparison in Table 1 was about design and this one is about evidence. Table 23 sets it out. The instructive column is the last: on magnitude, the dimension a multiplier estimate needs most, the four methods fail differently rather than uniformly. Fritsch and Mazlish (2026) do recover percent-of-GDP magnitudes and validate them against the Romer and Romer series, but for one country and for an expectations series rather than realised liabilities. Das et al. (2026) decline magnitude outright and are explicit about it. Bhasin and Loungani (2026) size their episodes and, where they diverge from the benchmark, argue the benchmark is the inflated one rather than conceding an accuracy limit, which is a defensible position and an unresolved one. Ours are extracted from the announcing document and traceable to a quoted rate, and carry no validation record at all. No method in this literature yet has a magnitude series that is both externally validated and portable.

Table 23: What the evidence in each method supports. The complement of Table 1 : not what each design does, but what its validation licenses and what it does not. Read alongside Table 22 , which does the same job stage by stage for this paper.

Method	Validation evidence	Scored against	Not supported by that evidence
Das et al.	Concordance on overlapping country-years	Action-based consolidation datasets	Magnitudes; countries with no benchmark series
Bhasin et al.	Divergence from benchmark decomposed into sizing and event selection	OECD action-based benchmark	Transfer beyond the OECD; which of the two series is closer to the truth
Fritsch et al.	High correlation with the narrative tax-shock series	Romer and Romer, United States	Realised liabilities (the series is expectations); any non-US setting
This paper	Behavioural, zero-shot and manual error analysis on 44 US acts; cross-language consistency; top-rate concordance with the Végh and Vuletin rate panel	Romer and Romer labels; Végh and Vuletin Malaysian CIT and PIT statutory rates; no Malaysian motivation ground truth	Magnitudes beyond the headline top rate; quarterly timing; the exogeneity column pending expert adjudication

7 A first use case: procyclicality decomposition

This section is a demonstration, not an inferential claim. It exists to show what a motivation-coded inventory lets a researcher ask that a statutory-rate series does not, and the question is the one Kaminsky et al. (2004) opened and Frankel et al. (2013) sharpened: when a country's fiscal record looks procyclical, is that deliberate response to the cycle, or the accumulation of measures taken for unrelated reasons that happened to land badly? A motivation-coded inventory is the only kind of dataset that can tell those apart, which is why Vegh and Vuletin (2015) could document the pattern and explicitly decline to attribute it.

7.1 What the statutory-rate measure can see

The standard measure of tax-policy cyclicality correlates the change in a statutory rate with the cyclical component of output. Run on Malaysia it returns nothing: 0.02 for the corporate rate and 0.01 for the personal rate, neither remotely distinguishable from zero.

The reason is not that Malaysian tax policy is acyclical. It is that a headline statutory rate moves rarely, and a correlation computed on 9 corporate rate changes in 44 years has almost no information in it. This is not a Malaysian peculiarity. Figure 19 computes the same statistic for every country in the panel with enough data, and the median one changed its corporate rate 6 times in four decades. The measure is thin nearly everywhere, and Malaysia sits close to the middle of the resulting spread.

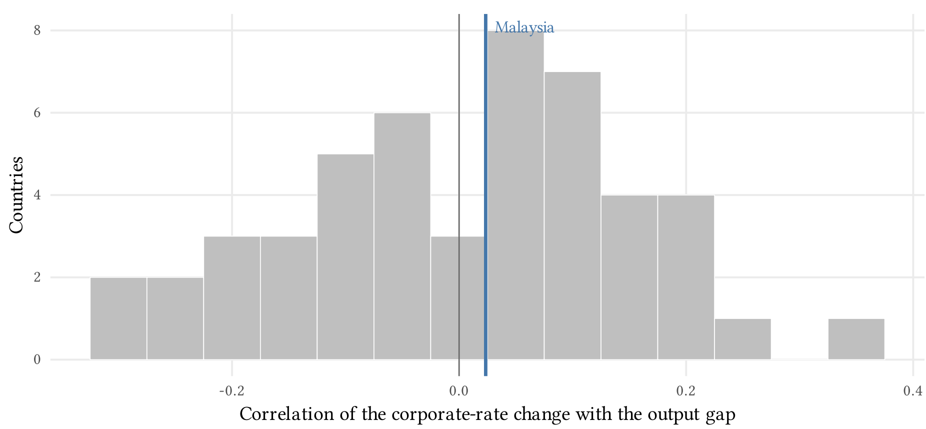

Figure 19: The statutory-rate measure is thin nearly everywhere. Distribution across countries of the correlation between the annual change in the corporate statutory rate and the output gap. Malaysia is marked. The spread is centred near zero and is dominated by how few times a headline rate moves.

Two things follow. The measure is under-powered, which more data would fix. And it is silent on attribution, which more data would not: a correlation between rate changes and the cycle cannot distinguish a rate cut enacted *because* output was strong from one enacted for unrelated reasons that happened to arrive in a boom. That second limitation is the one the narrative inventory addresses.

7.2 The fiscal record against the cycle

Figure 20 lays the inventory over Malaysia's business cycle. The upper panel is the output gap; the lower panel is one mark per fiscal action, placed at the year it took effect, above the line if it was expansionary and below if contractionary, coloured by whether the codebook judged its motivation exogenous.

Note that the signing axis here is the **fiscal impulse**, not the direction of the change in liabilities. A spending increase and a tax cut are both expansionary; the inventory figures earlier in the paper sign by liability direction, which is the right axis for describing a tax system and the wrong one for describing a stance.

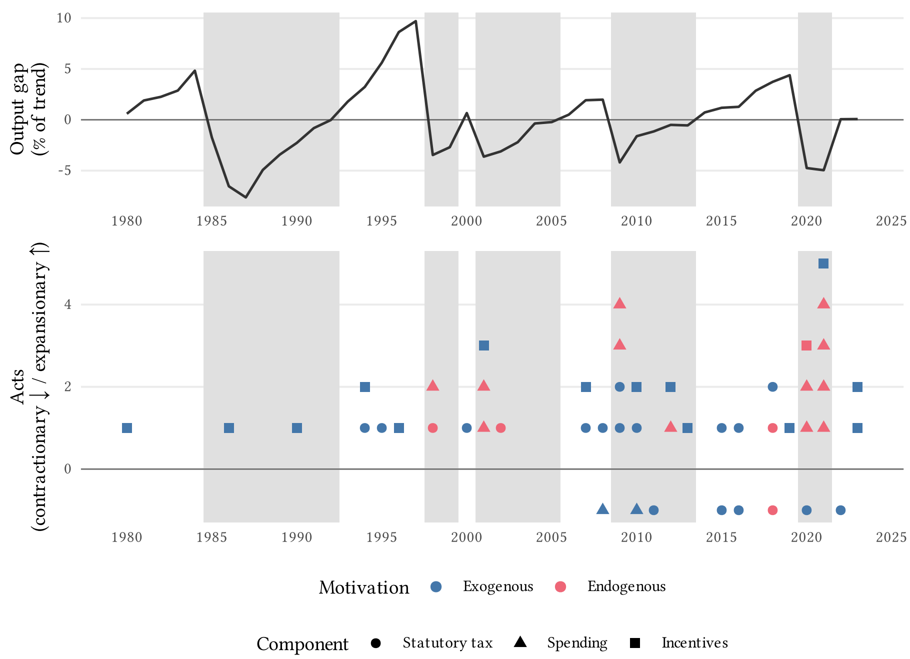

Figure 20: The Malaysian fiscal record against the cycle. Shaded bands are below-trend years, drawn behind both panels so a reader can drop from a downturn straight into the actions taken during it. Marks are stacked within a year rather than jittered, since the year is the claim.

7.3 The decomposition

Crossing the impulse with the cycle state gives each action a **stance**. An action is procyclical when it pushes the way the cycle is already going: expansionary in a boom, contractionary in a bust. Crossing stance with the motivation label gives Table 24 , and with it the quantity Vegh and Vuletin (2015) named as missing.

Of the 49 actions that carry both a stance and a motivation, 19 look procyclical on timing. **18 of them carried a motivation unrelated to the cycle.** Malaysia's procyclical-looking fiscal record is, on this reading, overwhelmingly not a record of deliberate procyclical policy. It is long-run and deficit-driven measures that happened to land against the cycle.

Table 24: The decomposition. Upper block: stance against motivation, all components. Lower block: the procyclical row under alternative cycle definitions, event timing, and component scope. The reported p is Fisher's exact test, which for a 2×2 with both margins fixed *is* the permutation test.

Stance / specification	Exogenous	Endogenous	Total	Exact p
Countercyclical	14	16	30	
Procyclical	18	1	19	0.00057
Growth below within-country median	16	1	17	0.0019
Dated by announcement year	19	3	22	0.0059
Excluding spending	17	1	18	0.18

Figure 21 shows where each column's mass comes from, and makes the split legible: the countercyclical column is dominated by acts the codebook labelled countercyclical, which is close to a definitional result. The procyclical column is long-run and deficit-driven measures, which is not.

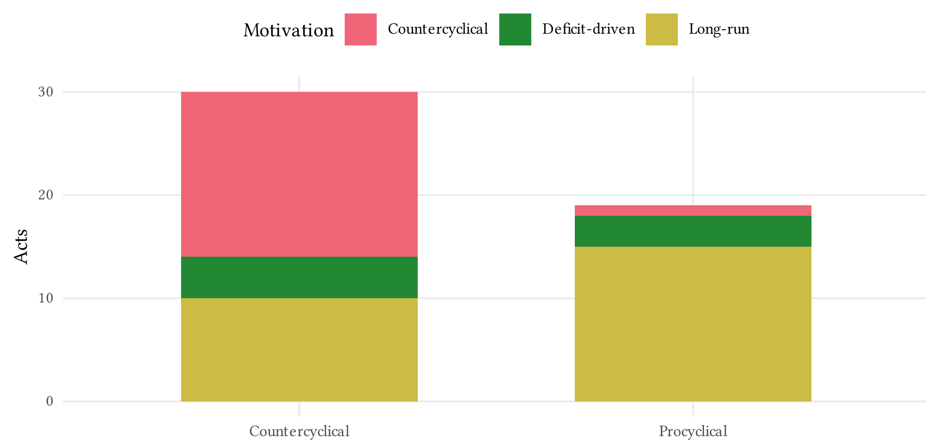

Figure 21: Stance by motivation. The countercyclical column is mostly acts labelled countercyclical, which is near-definitional. The procyclical column is not, and it is the one the decomposition is about.

Table 25 sets the two measures side by side. The point is not that the narrative decomposition is more precise than the correlation. It is that they answer different questions, and only one of them is the question about attribution.

Table 25: What each measure supports. The Vegh and Vuletin (2015) statutory-rate statistic and the narrative decomposition, on the same country and the same years.

Measure	Value
Végh–Vuletin rate cyclicality, corporate	0.02 (p = 0.88)
Végh–Vuletin rate cyclicality, personal	0.01 (p = 0.97)
Rate changes available to it	9 corporate, 11 personal, in 44 years
Rate changes, median country in the panel	6 across 49 countries
Narrative decomposition, procyclical acts	19 of 49 acts
Of those, motivated by something other than the cycle	18 (exact p = 0.00057)
Relabels needed to overturn it	4 of 18; 2 already flagged by the pipeline

7.4 How much to believe it

Three reasons for caution, in descending order of how much they should worry a reader.

The labels are model output, and a p-value cannot speak to that. At this sample size the binding uncertainty is not sampling noise. So the quantity we report instead is a **fragility index**: the number of exogenous acts in the procyclical cell that would have to be relabelled endogenous before the association stopped being significant. It is 4, out of 18. Set against that, the pipeline already flags 2 of the procyclical acts as cases where its two independent reads disagree — below the breaking point, but not by much. This is also the concrete cost of the defect in Section 4.5: four of the 51 motivation labels are known to be wrong in these exhibits, the same order of magnitude as the fragility index, which is why the correction is a prerequisite for the estimate rather than a tidying-up exercise.

The result thins when spending is removed. The last row of Table 24 drops the spending component and the exact test stops being significant. That is expected and worth stating plainly: the stimulus packages are both endogenous and countercyclical, so they supply most of the contrast. What survives without them is the point estimate rather than the test (17 of the 18 procyclical tax and incentive actions still carry an exogenous motivation), and at this sample size a point estimate is the honest object anyway.

Generated regressors. Once an estimate is built on the labels, valid inference is a measurement-error problem, not a classification-accuracy one (Ludwig et al. 2025; Battaglia et al. 2024). The bias equals the correlation of label error with the regressor, and the dangerous regressor is the business cycle, the exact axis exogeneity turns on. So a downstream estimate must either show mislabelling is uncorrelated with the cycle or scope the claim accordingly, and would properly debias against a probability-sampled expert validation set (Angelopoulos et al. 2023; Egami et al. 2023). The expert protocol in Section 5.2 already routes every act predicted exogenous in a downturn to full review, which is exactly the fragility set; Section 6 notes that it needs a probability sample as well as a purposive one.

One supporting diagnostic. Shifting the cycle series circularly through every lag, which preserves its serial correlation and the events' clustering in time, puts the observed association ahead of all 43 shifted alternatives. Only as many distinct shifts exist as there are years, so this is a coarse check rather than a second test, and it is reported as one.

What is deliberately not here. The natural next estimator is a local projection of output on the signed exogenous events (Jordà 2005). We do not run one. It would need the magnitude column, and Table 22 records that magnitudes carry no validation record beyond the headline statutory rate; an impulse response built on an unvalidated regressor would look like the paper's main result while being its weakest. It is the first thing a validated magnitude codebook would unlock, and it is the reason Section 6 treats magnitude as a binding gap rather than a refinement.

8 Conclusion

The introduction set out four questions. This paper settles two of them in part, leaves one alone, and delivers the fourth for one country.

On **Conjecture** `(..) => ..` : **an LLM can reproduce narrative identification against Romer and Romer's own labels for three of the five quantities the method recovers.** Which passages describe a fiscal measure, what motivated each act and whether that motivation was independent of current output, and the direction of the change: each carries a full behavioural, zero-shot and error-analysis record. Magnitude in percent of GDP and quarterly timing do not, and we have been careful not to let the validated parts lend credibility to the unvalidated ones. The most useful thing we learned here was not a score. It was that a class of apparent model errors turned out to be errors in the benchmark, which is a finding about evaluating codebooks against projected human labels that any project at this scale will run into.

On **Conjecture** `(..) => ..` : **the evidence that frozen codebooks transfer to a country with no labelled data is favourable, and it is not an accuracy rate.** It cannot be, because the absence of ground truth is the condition the method exists to work under rather than an obstacle on the way to better conditions. What we have instead is a bilingual corpus returning comparable inventories in both languages, a set of disagreement cases printed in full for the reader to judge, and statutory rate paths that reproduce the Vegh and Vuletin (2015) cross-country rate panel exactly wherever the corpus is dense. None of the three is sufficient, and the one that would be, expert adjudication, is running now. There is also a result here that cuts against the paper's own interest: the ablation in Section 5.1.2 suggests the classifier is running substantially on a pretrained prior rather than on our codebook, which is why transfer is plausible and equally why demonstrating it on US-adjacent material would prove little.

On **Conjecture** `(..) => ..` : **we have nothing to report on comparable cross-country multipliers, and the reason is now specific.** It needs magnitude, quarterly timing, and more than one country. Section 7 declines to run the local projection that would be the natural first step, because the magnitude column has no validation record beyond the headline statutory rate. That is not a caution about ambition; it is the concrete price of the next codebook.

On **Conjecture** `(..) => ..` : **the Malaysia inventory is a dataset a local researcher can use, and its provenance chain is the reason.** Every event traces to a quoted passage in a named primary-source document. That is what let a reviewer's objection to two corporate-tax rows surface a machinery defect that had silently mislabelled events across all three components (Section 4.5), and it is what makes Section 7's decomposition checkable rather than merely reported. A dataset whose errors can be found is worth more than one whose errors are invisible.

The use case in Section 7 is the clearest statement of what the exercise buys. Malaysia's statutory-rate cyclicality is indistinguishable from zero, and so is the median country's, because a headline rate moves a handful of times in four decades and a correlation on those few movements cannot separate policy responding to the cycle from policy that merely arrived during one. The narrative inventory answers that second question directly, and its answer for Malaysia is that the procyclical-looking record is overwhelmingly *not* deliberate procyclical policy. We report that as a demonstration of what the data supports, hedged by a fragility index rather than a p-value, because the labels are model output and four relabellings would overturn it.

Which is the honest summary of the whole paper. The claim is not that the pipeline replaces the expert. It is that it changes what the expert spends months doing, and that the resulting dataset carries, alongside every number, an account of how much weight that number will bear.

9 Limitations and roadmap

Table 22 states what is and is not evidenced. Three further limitations are about the design rather than its current state.

Transfer is a claim to defend, not to assume. Fixed text measures notoriously fail to transfer across domains (Gentzkow et al. 2019), and cross-codebook fine-tuning overfits (Halterman and Keith 2025), so frozen country-agnostic codebooks have to be shown to survive contact with a new country rather than assumed to. The cross-language test and the regional extension are what put that claim at risk. The finding in Section 5.1.2 that the classifier appears to be running on a pretrained prior cuts both ways here: it is why transfer is plausible, and it is why demonstrating transfer on US-adjacent material proves little.

One country is one observation. Malaysia was chosen for a stable political window, a bilingual archive, and a documented fiscal record. Indonesia, Thailand, the Philippines, and Vietnam follow the same pipeline, and the parts of the method that turn out to be Malaysia-specific will only be visible then.

The human stamp is load-bearing and does not scale linearly. Act identification and consolidation require a researcher to confirm the events and the grain, per instrument, per country. The pipeline reduces months of work to weeks; it does not remove the expert, and the claim in this paper is explicitly about assistance, not automation.

Appendix

A Recall audits

The per-instrument recall audits, frozen with their datasets. As Section 4.3 says, these record what the identification pass searched for and what it returned; they are a transparency record, not a recall measurement, and the recall assessment that matters is the expert review described in Section 5.2. Rows naming a recall miss record an event the codebook scan did not surface and the direct-reading pass recovered. Outcome text is truncated for display.

Table A.1: Recall audit, corporate income tax.

Stage	Outcome
Keyword scan (C1 pool, n=1,036)	29 keyword hits over CIT regex; 2 false positives excluded (Malaysia Deposit Insurance Corporation Act 2005 x2).
Semantic sweep (all surface forms)	Confirmed headline path 34->24 plus separate SME / petroleum / sectoral-incentive tracks (tracked separately, not headline rows).
C1 measure identification	Surfaced the 34->25 path (steps H1-H6) and the single-tier reform; MISSED the 2016 cut (25->24) and the 2022 Cukai Makmur surcharge.
Recall miss - 2016 cut (25->24, YA2016)	Recovered from Budget Speech 2015 ch5 (BM: 'daripada 25 peratus kepada 24 peratus') and Economic Report 2016 ch13 ('Corporate income tax rate reduced from 25% to 24%').
Recall miss - 2022 Cukai Makmur (YA2022)	Recovered from Budget Speech 2022 ch18 ('one-off special tax known as Cukai Makmur … companies other than MSMEs … above RM100 million … 33% tax rate') and Economic Report 2023 ch7-8 ('Prosperity Tax amou…
C0 act aggregation	Fragmented the CIT measures (near-all singletons; phased cut split across approx. 10 clusters; EN/BM never bridged). NOT used as the event layer.

Table A.2: Recall audit, personal income tax.

Stage	Outcome
Keyword scan (C1 pool, n=1,036)	30 hits over the PIT regex spanning 1999-2023; approx. 2 false positives excluded (a 2004 corporate 20%-band item; a SOCSO taxi-driver relief item).
Semantic sweep (all income-tax surface forms)	Confirmed the top-marginal-rate path 30->29->28->27->26->25 then back up to 28 (YA2016) and 30 (YA2020); relief/rebate/threshold tracks recorded as magnitude detail, not headline rows.
C1 measure identification	Surfaced P1, P2, P4, P5, P7, P8, P9; MISSED P3 (28->27, Budget 2009) and P6 (25->28 high-earner hike, Budget 2016).
Recall miss - Budget 2009 (28->27, YA2009)	Recovered from Budget Speech 2009 ch2 (BM: 'kadar cukai individu maksimum diturunkan dari 28 peratus kepada 27 peratus, mulai tahun taksiran 2009'); retrospectively confirmed in Budget Speech 2010 ch6.
Recall miss - Budget 2016 high-earner hike (25->28, YA2016)	Recovered from Budget Speech 2016 ch3/9 + Appendix 1 ('chargeable income exceeding RM1,000,000 be increased by 3 percentage points … 25% … 28%').
C0 act aggregation	Not used as the event layer (same fragmentation finding as CIT: near-all singletons; EN/BM never bridged).

Table A.3: Recall audit, consumption tax (GST/SST).

Stage	Outcome
Keyword scan (C1 pool, n=1,036)	93 keyword hits over the consumption-tax regex; GST dominates (44 hits), SST/sales-tax/service-tax the remainder.
Semantic sweep (all surface forms + source docs)	Confirmed four headline statutory events: service tax 5->6 (2011), GST introduction at 6% (2015), GST abolition (2018), SST reintroduction (2018). No further broad-rate change found 1980-2023.
C1 measure identification	Surfaced all four events as measures (good event-level recall, unlike CIT).
Recall miss - SST reintro rate detail (YA2018)	Sales Tax (5%/10%) & Service Tax 6% rate detail (ECON Report 2019 ch12) and SST-effective-date passages (BNM AR 2018 ch6/16) not surfaced; recovered by reading the source docs.
Recall miss - service-tax corroboration (2010)	BNM Annual Report 2010 service-tax quote absent from country_chunks (doc not chunked); recovered as recovered_evidence.
Pre-corpus / out-of-scope tracks	Product-specific sales-tax changes (liquor/cigarettes 15->20/25% Budget 2001; 1997/98 crisis exemptions) and GST base/exemption tweaks (2016/2017) tracked but excluded as narrow carve-outs, not standard-rate s...
C0 act aggregation	Not used as the event layer (fragments GST across many clusters; EN/BM not bridged), consistent with the CIT/PIT passes.

Table A.4: Recall audit, spending.

Stage	Outcome
C1 keyword scan (C1 pool, n=1,036)	NOT near-empty: 189 spending-regex hits. BNM Annual Reports, Economic Reports and Budget Speeches discuss stimulus packages and subsidy policy alongside tax, so C1 surfaced most spending events. Deviates from ...
Direct corpus sweep (country_body / country_chunks)	Read Development-Expenditure / crisis-booklet / Economic-Report / BNM-AR sections directly. Confirmed C1-surfaced events; recovered_chunks empty for every act (events were already C1-visible, the opposite of t...
Recall checkpoint - 1998 NERP	FOUND. National Economic Recovery Plan (Asian-crisis response): MY_NERP-1998, MY_BNM_AR-1998, MY_ECON_REPORT-1999, MY_BUDGET_SPEECH-1999. -> MY-SPEND-01.
Recall checkpoint - 2009 GFC stimulus	FOUND. First Economic Stimulus Package (RM7bn, Nov 2008) + Second Economic Stimulus Package (RM60bn, Mar 2009 mini-budget) = RM67bn: MY_BNM_AR-2008/2009, MY_MINI_BUDGET-2009, MY_ECON_REPORT-2010. -> MY-SPEND-0...
Recall checkpoint - 2020/21 COVID (PRIHATIN/PENJANA/PEMERKASA)	FOUND. 8 packages, RM530bn+ (NRP-2021). Per-package grain (reviewer decision): PRIHATIN, PENJANA, PERMAI, PEMERKASA, PEMERKASA+, PEMULIH each a row; KITA PRIHATIN / PRIHATIN SME+ folded into PRIHATIN. -> MY-SP...
Consolidation grain (reviewer adjudication)	RMK Five-Year Plans EXCLUDED (baseline development-expenditure framework, not discrete shocks). COVID split per named package. 2008 fuel-subsidy restructuring kept as a separate (endogenous) row distinct from ...
C0 act aggregation	Not used as the event layer (C0 is tax-scoped; fragments tight spending tracks and does not bridge EN/BM). Consolidation done by direct reading.

Table A.5: Recall audit, tax incentives.

Stage	Outcome
Keyword scan (C1 pool, n=1,036)	NOT near-empty (skill expected near-empty): 151 measures matched the incentive regex, 1981-2023, EN+BM. Malaysia Budget/ER/BNM/Plan docs discuss incentives heavily, so C1 surfaced them directly.
Seed PREFERENTIAL_RATE recovery (CIT carve-outs)	All present: DEB 5pp (1981), Labuan/LOBATA (1990), biomass (2001), Principal Hub 10% (2019), pharma/vaccine 0-10% (2021), relocation 0/10/15% (2020/2023).
Framework instruments (PIA / RA / MSC)	All present: PIA 1986 enabling reform, Reinvestment Allowance 40%->50% (Budget 1994), MSC Bill-of-Guarantees Pioneer Status.
Major named regimes (reviewer breadth decision)	Added Iskandar/IDR (2007), GBI green-building (2010), KLIFD/TRX (2012), angel-investor & VC (2013), EV (2023).
recovered_chunks	Empty for every act – C1 surfaced the incentives directly (like spending). Direct reading served consolidation/rating, not recovery.
recovered_evidence	One synthetic quote for Iskandar/IDR (MY-INCENT-07): corpus records the corridor launch + investment fund, not the precise IDR tax-exemption schedule.
Long tail NOT rowed	approx. 30+ narrower annual sectoral incentives left out per reviewer grain decision (e.g. REITs, Islamic finance, OHQ, childcare, Industry4WRD, asset-backed securities, R&D double-deduction, tourism, agricult...

B How the US gold labels were built

Romer and Romer publish act-level classifications. The measure-identification codebook reads chunks of documents. Evaluating one against the other requires deciding, for every chunk in the US corpus, whether it is a positive: whether a known act is substantively discussed in it. There is no published answer to that question, so it has to be constructed, and this appendix records how, because Section 5.1.1.4 shows the construction is the largest single source of error in the US benchmark.

Two obvious approaches fail. Hand-coding twelve thousand chunks is infeasible. Exact string matching of act names fails for a more interesting reason: government PDFs from the 1940s to the 1990s were digitised by OCR, which routinely breaks an act name across a line, so "Revenue Act of 1962" appears in the extracted text as `Revenue Act of\n1962` and matches nothing. Normalising whitespace recovers a large number of matches, and for several acts it doubles the count, but it does not rescue acts that are never named in the form the reference list uses.

The construction therefore assigns chunks to two tiers of positive, plus a negative pool.

Tier 1 chunks contain an act's own narrative passage verbatim. Romer and Romer publish, with each act, the passages of presidential messages and committee testimony that document its motivation. Searching the corpus for the first 100 characters of each passage, with a 60-character fallback, locates the chunk that passage came from. These are as close to certain positives as the construction reaches, and there are few of them, because one act contributes only as many Tier 1 chunks as it has published passages.

Tier 2 chunks name a known act without containing its passage. Because act names are unreliable strings, three matching strategies are used: whitespace- normalised substring search; decomposition of compound names into searchable subcomponents, which is what makes "Revenue Act of 1962" findable inside "Expiration of Excess Profits Tax and of Temporary Income Tax Increases"; and a small set of hand-written co-occurrence rules for acts that no substring reaches. Deliberately, no year window and no keyword filter is applied. The tier is built to maximise recall on the theory that downstream stages can filter.

Negatives are every remaining chunk, scored with a continuous relevance-key density rather than excluded by a threshold. An earlier design excluded any chunk containing fiscal vocabulary, which left 35 clean negatives out of thousands and made precision essentially untestable.

The resulting split is heavily imbalanced: on the US corpus, Tier 1 accounts for 0.6% of chunks, Tier 2 for 12.2%, and negatives for 90.4%.

The weakness is in Tier 2, and it is structural rather than a tuning problem. An act *name* appears in many places its substance does not: book indexes, tables of contents, glossaries, running heads, and administrative appendices listing every statute touched. Those chunks enter the evaluation carrying a positive gold label and containing no fiscal content at all. When the codebook correctly declines to call a table of contents a fiscal measure, the evaluation records a false negative. That is the mechanism behind the Category B count in Section 5.1.1.4, and it is why the corrected gate, rather than the raw recall figure, is the number to read.

C Codebooks

The codebooks are reproduced verbatim, as the model would read them.

C.1 Measure identification

```
codebook:
  name: "C1: Fiscal Measure Identification"
  version: "0.7.0"
  instructions: >
    You are an expert in fiscal policy analysis classifying chunks from official
government
     economic and budget documents. Your task is to scan the entire chunk and
determine whether
     **any portion** of it substantively discusses a specific fiscal measure —
enacted, proposed,
```

```
     or under consideration — that changes or would change fiscal liabilities
or obligations.
   Chunks from government documents typically contain extended economic analysis,
forecasts,
     and policy discussion — a chunk should be classified as FISCAL_MEASURE if
it contains even
    one substantive discussion of a specific fiscal measure, regardless of what
the rest of the
    chunk discusses. Read the class definitions carefully and classify the chunk
using only the
    provided labels.

   The chunk you are analysing comes from a corpus published by the government of
   **{country}** (ISO code: `{country_iso}`). Some chunks discuss fiscal measures
enacted by other countries
       (foreign comparators, supranational bodies); these still qualify as
FISCAL_MEASURE
      because they describe fiscal policy, but their country-of-enactment is
`OTHER`, not
    `{country_iso}`. The country-of-enactment tag is applied per measure, since
a single
    chunk may discuss measures from multiple countries.

  classes:
    - label: "FISCAL_MEASURE"
      label_definition: >
      A chunk that contains, in any portion of its text, a substantive discussion
of a specific
        legislative or executive action — whether enacted, proposed, or under
consideration — that
        changes or would change fiscal liabilities or obligations.

      clarification:
       - "The passage describes a change in fiscal liabilities from taxes of any
type — income taxes, payroll and social insurance taxes, consumption and excise
taxes, investment incentives, and so on"
          - "The action may be enacted into law, implemented by executive or
administrative action (such as executive orders, ministerial decrees, regulatory
changes, or official policy directives), proposed in legislation, recommended by
officials or advisory bodies, or under consideration; C1 captures all substantive
discussion of specific fiscal measures regardless of legislative status"
      - "Retrospective references that name a specific act and provide substantive
detail about its provisions — such as rate changes, affected populations, or
```

```
revenue effects — should be classified as FISCAL_MEASURE, even if the passage
was written years after enactment"
          - "The passage names a specific act, proposal, or executive action
and provides some indication of its fiscal implications — this could include
rate changes, affected populations, revenue estimates, effective dates, program
descriptions, or even a general description of the measure's purpose and scope —
rather than merely naming the measure or describing only its political history
without indicating what it would change"
          - "The significance of a measure is determined by how substantively
the passage discusses it, not by its revenue impact; even small fiscal changes
qualify if described in detail"
          - "Foreign-government fiscal measures (e.g., a chunk from a {country}
government document that discusses a Japanese consumption-tax change or a U.S.
stimulus package as a comparator) still qualify as FISCAL_MEASURE; tag the per-
measure country as `OTHER`"

      negative_clarification:
        - "Simple extensions or renewals of existing provisions that do not alter
rates, coverage, or scope are excluded from classification as fiscal measures"
        - "Changes to withholding schedules, payment timing, or tax collection
procedures that do not alter the underlying liabilities owed by taxpayers or
recipients are excluded"
        - "Automatic adjustments such as inflation indexing that occur mechanically
without new legislative action are excluded"
          - "Passages that consist entirely of tabular data listing multiple
fiscal actions with only brief numerical entries — such as tables of revenue
effects with no accompanying narrative — are excluded; however, if any part of
the passage provides a narrative description of what a specific measure changed,
the chunk qualifies as FISCAL_MEASURE"
        - "Government financial actions that do not change liabilities owed by
or owed to taxpayers are excluded — this includes intergovernmental transfers
(e.g., revenue sharing between levels of government), debt management operations
(e.g., issuing new types of securities), organizational or administrative
restructuring of government agencies, and spending authorizations that set
program funding levels without altering tax obligations or direct transfer
payments to individuals or firms"

    - label: "NOT_FISCAL_MEASURE"
      label_definition: >
        A chunk in which no portion of the text substantively discusses a specific
fiscal measure
        that changes or would change fiscal liabilities or obligations.
```

```
      clarification:
        - "General economic analysis, forecasts, or trend discussion that does not, anywhere in the chunk, reference a specific fiscal action with enough detail to understand what it changed or would change"
        - "Discussion of fiscal aggregates such as total revenue, total spending, or deficit levels without identifying the specific legislative or executive measure causing the change"
        - "Administrative guidance, regulatory procedures, tax collection processes, or compliance reporting requirements that do not alter underlying fiscal liabilities"
        - "Passages that merely name past legislation without providing substantive detail about what the measure changed (e.g., 'following the reforms of the 1980s') are not fiscal measures"

      negative_clarification:
        - "A passage is not excluded from FISCAL_MEASURE classification merely because it contains fiscal vocabulary such as 'tax', 'revenue', 'spending', or 'budget'; it must describe a specific fiscal measure to qualify"
        - "Brief historical references that merely name past legislation without substantive detail about provisions are NOT_FISCAL_MEASURE, but retrospective descriptions that provide substantive detail about an act's specific provisions should be classified as FISCAL_MEASURE"

  output_instructions: >
    Classify the chunk using exactly one of: FISCAL_MEASURE, NOT_FISCAL_MEASURE.

    If FISCAL_MEASURE, return the list of all distinct fiscal measures discussed in the chunk
     in the `measures` array. **List all measures discussed; the first array element must be
     the most prominent.** A chunk legitimately may discuss two or more acts (for example,
     "unlike Act X, which was countercyclical, Act Y was long-run") — list each as a separate
     `measures[]` entry so downstream analysis can attribute evidence to the correct act.
    Empty `measures: []` is invalid for FISCAL_MEASURE labels.

     **One act = one entry.** If a chunk discusses multiple provisions of a single named
    act (e.g., a rate-reduction provision and a rebate provision of the same tax act, or
    several sections of one omnibus bill referred to by the omnibus name), list
```

```
that act
    **once** with a single `measures[]` entry; do not split sub-provisions of
the same act
    into separate entries. Two entries are warranted only when the chunk names
two (or
    more) distinct acts.

    For each measure, populate five per-measure fields:

    - `name`: the act name, proposal name, or executive-action descriptor as it
appears in
      the chunk (preserve the chunk's wording rather than canonicalising)

    - `country`: `{country_iso}` if the measure was enacted (or proposed) by
the {country_iso}
      government; `OTHER` if it is a foreign comparator, a measure of a
supranational body
      (IMF programs, EU directives, ASEAN agreements, etc. — always `OTHER`
regardless of
      which member country implements them), or otherwise enacted by a different
country.
      The only two legal values are `{country_iso}` and `OTHER`.

    - `discusses_motivation`: does the chunk discuss **why** this measure was
enacted — its
      purpose, stated goals, or the economic/political conditions that prompted it?

    - `discusses_timing`: does the chunk discuss **when** this measure takes or
took effect —
      implementation dates, phase-in schedules, or retroactive provisions?

    - `discusses_magnitude`: does the chunk discuss **how large** this measure
is — revenue
      estimates, dollar amounts, rate changes, or affected population sizes?

    If NOT_FISCAL_MEASURE, return `measures: []` (empty array).

    Return your answer as JSON:

    {
      "label": "FISCAL_MEASURE" or "NOT_FISCAL_MEASURE",
      "measures": [
        {
```

```
          "name": "Name of the fiscal measure",
          "country": "{country_iso}" or "OTHER",
          "discusses_motivation": true or false,
          "discusses_timing": true or false,
          "discusses_magnitude": true or false
        }
      ],
      "reasoning": "Brief explanation citing specific evidence from the chunk"
    }
```

C.2 Motivation evidence extraction

```
codebook:
  name: "C2a: Motivation, Enacted-Status, and Timing Evidence Extraction"
  version: "0.5.1"
  instructions: >
    You are an expert in fiscal policy analysis extracting evidence from chunks of
      official government economic and budget documents. Each chunk has been
identified
      as containing a substantive discussion of a fiscal measure, and you are
told which
      measure to work on in the `Act:` field of the input. A chunk may discuss
**several
    distinct fiscal measures** — budget speeches in particular often enumerate many
      unrelated measures in sequence — so your task is to extract evidence
about **the
    named measure only**, not about everything the chunk happens to mention.

    Your task is to extract three types of evidence about the named measure:

    1. **Motivation evidence**: All evidence about **why** the fiscal measure was
      enacted, proposed, or under consideration — the stated goals, economic
conditions
      cited as justification, and any other indications of the policymakers'
motivation.

    2. **Enacted-status signals**: Any evidence about the **legislative status**
of the
      measure — whether it was enacted into law, took legal effect (including
via executive
    or administrative action), was defeated, withdrawn, or remained a proposal.
These
    signals will be used downstream to determine whether the measure became law.
```

```
   3. **Timing signals**: Any evidence about **when** the measure changed fiscal
     liabilities. This includes effective dates ("effective January 1, 1987",
"took effect
    July 1, 1950"), signing dates ("signed into law on October 22, 1986" — used
as a
    fallback timing cue when no explicit effective date is stated), retroactive
language
    ("retroactive to January 1948", "applied retroactively to..."), and phased
      implementation language ("rates will increase in 1954, 1960, 1965, and
1970", "first
      phase begins..."). Quote each timing-bearing fragment separately. **Pay
particular
      attention to language about retroactivity to January 1 of the current
calendar year
      or to a date earlier in the calendar year than the signing date** —
these effective
    dates often appear separately from (and sometimes far from) the signing date
in the
    text, and they define the actual implementation quarter under R&R's standard
series.
     Extract the retroactive effective date as a distinct timing signal even
when the
     signing date is also in the chunk. The downstream classifier will apply
Romer &
   Romer's midpoint rule (effective on or before quarter midpoint → that quarter;
     strictly after midpoint → next quarter) and treat retroactive components
per the
   standard series (implementation quarter only, no separate retroactive entry).

    You will be given: the chunk text, the name of the fiscal measure, and the
year of
     passage. For each distinct piece of evidence (motivation, enacted-status,
or timing),
     extract a direct quote and describe the signal it represents. The same
quote may
     appear in more than one array if it serves more than one role (e.g., a
signing date
    is both an enacted-status signal and a fallback timing signal).

     **Measure scope.** Evidence counts only if it bears on the measure named
in the
     `Act:` field. When a chunk describes several measures, extract evidence
```

```
for the
   named one and leave the others alone, however well-stated their rationales are.
    Two common cases to get right:

      - A chunk enumerates many measures (a budget speech listing reliefs,
allocations,
     and rate changes in sequence). Extract only the quotes that state why
**the named
    measure** was taken. Rationales attached to the neighbouring measures are
out of
    scope even when they are the most quotable lines in the chunk.

   - A chunk states a broad fiscal-policy stance (fiscal consolidation, a revenue
    strategy, a reform programme) alongside the named measure. Such framing is
in scope
   **only** when the text ties it to the named measure as its rationale. General
   policy background that the chunk does not connect to the named measure is not
    evidence about it.

   Scoping does not mean demanding an exact name match. When the chunk discusses a
   single measure, that measure **is** the named one even if the chunk's wording
   differs from the `Act:` string (an abbreviation, a translation, a descriptive
     paraphrase, or a different level of detail). Match on substance, not
on wording.
    Return an empty `evidence` array only when the chunk genuinely says nothing
about
    why the named measure was taken.

     Extract only evidence that is explicitly present in the text. Do not
infer dates,
     motivation, or status that are not stated. If a chunk has no timing-
bearing language,
    return an empty `timing_signals` array.

  examples:
    - input:
        act_name: "Excess Profits Tax Act"
        year: 1950
        text: >
            The Excess Profits Tax Act, passed in response to the Korean War,
imposed a
        heavy tax on corporate excess profits to finance the war effort. Although
           signed into law on January 3, 1951, the act was made retroactive to
```

```yaml
July 1,
            1950. The tax was expected to raise revenues by $3.5 billion at
an annual
          rate. The retroactive component covered two quarters of profits that had
          already been earned, resulting in a one-time levy in the first quarter of
            1951 in addition to the ongoing annual rate.
      output:
        evidence:
          - quote: "passed in response to the Korean War, imposed a heavy tax
on corporate excess profits to finance the war effort"
        signal: "Revenue measure motivated by financing wartime defense spending"
        enacted_signals:
          - quote: "signed into law on January 3, 1951"
            signal: "Signed into law"
        timing_signals:
          - quote: "signed into law on January 3, 1951"
          signal: "Signing date — January 3, 1951 (1951Q1); fallback timing cue"
          - quote: "the act was made retroactive to July 1, 1950"
              signal: "Retroactive effective date — July 1, 1950 (1950Q3 under
midpoint rule)"
          - quote: "The retroactive component covered two quarters of profits
that had already been earned, resulting in a one-time levy in the first quarter
of 1951"
         signal: "Phased / retroactive structure: standard-series implementation
in 1951Q1; retroactive component covers 1950Q3–1950Q4 (excluded from standard
series)"

    - input:
        act_name: "Economic Recovery Tax Act"
        year: 1981
        text: >
          The Administration argued that high marginal tax rates were reducing
incentives
          for work, saving, and investment. The President stated that 'our tax
system has
             become an obstacle to growth, penalizing productive effort and
discouraging
            innovation.' By reducing marginal rates and indexing tax brackets
for inflation,
          the act aimed to raise long-run economic growth and improve economic
efficiency.
            The measure was enacted on August 13, 1981, with the first round
of rate
```

```
          reductions taking effect October 1, 1981, and additional cuts phased
in on
          July 1, 1982 and July 1, 1983.
      output:
        evidence:
          - quote: "high marginal tax rates were reducing incentives for work,
saving, and investment"
              signal: "Stated concern about structural disincentives from the
tax system"
          - quote: "our tax system has become an obstacle to growth, penalizing
productive effort and discouraging innovation"
              signal: "Stated goal of removing structural barriers to long-
run growth"
          - quote: "the act aimed to raise long-run economic growth and improve
economic efficiency"
        signal: "Stated goal of raising long-run growth and improving efficiency"
        enacted_signals:
          - quote: "The measure was enacted on August 13, 1981"
            signal: "Enacted into law"
        timing_signals:
          - quote: "enacted on August 13, 1981"
            signal: "Signing date — August 13, 1981 (1981Q3); fallback timing"
         - quote: "first round of rate reductions taking effect October 1, 1981"
              signal: "Phase 1 effective date — October 1, 1981 (1981Q4 under
midpoint rule)"
          - quote: "additional cuts phased in on July 1, 1982 and July 1, 1983"
            signal: "Phase 2 and Phase 3 effective dates — July 1, 1982 (1982Q3)
and July 1, 1983 (1983Q3)"

    - input:
        act_name: "Emergency Stimulus Act"
        year: 2009
        text: >
              With unemployment rising sharply and GDP contracting for the
second consecutive
           quarter, the government enacted temporary tax relief. The Secretary
of the Treasury
          testified that 'these measures are needed to counteract the recession
and restore
           the economy to its normal growth path.' The relief provisions were
designed to
          expire once economic conditions normalized.
      output:
```

```yaml
        evidence:
          - quote: "unemployment rising sharply and GDP contracting for the second consecutive quarter"
            signal: "Economic conditions cited as justification for the measure"
          - quote: "these measures are needed to counteract the recession and restore the economy to its normal growth path"
            signal: "Stated goal of counteracting recession and restoring normal growth"
          - quote: "The relief provisions were designed to expire once economic conditions normalized"
            signal: "Temporary measure tied to cyclical economic conditions"
        enacted_signals: []
        timing_signals: []

    - input:
        act_name: "Fiscal Responsibility Act"
        year: 1993
        text: >
          The deficit, accumulated over many years of past policy decisions, threatened
          long-term fiscal stability. The primary purpose of the act was to restore fiscal
          balance and reduce the structural budget deficit. The President signed the act
          into law on August 10, 1993, with provisions taking effect at the start of
          calendar year 1994.
      output:
        evidence:
          - quote: "The deficit, accumulated over many years of past policy decisions, threatened long-term fiscal stability"
            signal: "Deficit described as inherited from past decisions"
          - quote: "The primary purpose of the act was to restore fiscal balance and reduce the structural budget deficit"
            signal: "Stated goal of deficit reduction and fiscal prudence"
        enacted_signals:
          - quote: "The President signed the act into law on August 10, 1993"
            signal: "Signed into law"
        timing_signals:
          - quote: "The President signed the act into law on August 10, 1993"
            signal: "Signing date — August 10, 1993 (1993Q3); fallback timing"
          - quote: "provisions taking effect at the start of calendar year 1994"
            signal: "Effective date — January 1, 1994 (1994Q1)"
```

```yaml
# Multi-measure chunk. The chunk discusses two separable measures — a broad
# consumption tax and a corporate rate reduction — but only the corporate rate
# reduction is named. Evidence about the consumption tax is out of scope, however
# prominent it is in the text.
- input:
    act_name: "Corporate income tax rate reduction"
    year: 2016
    text: >
      Third: the national consumption tax will take effect in April, replacing the
      previous sales and services taxes. This reform broadens the revenue base and
      strengthens the fiscal position, and is central to the government's commitment
      to reduce the deficit and achieve a balanced budget by 2020. Following feedback
      from the public, the government has agreed to widen the list of basic food items
      exempted from the tax.

      Fourth: to keep our tax regime competitive with neighbouring economies and to
      encourage private firms to expand investment, the corporate income tax rate will
      be reduced by one percentage point, from 25 per cent to 24 per cent, with effect
      from the 2016 year of assessment.

      Fifth: the government will allocate additional funding to the small enterprise
      financing scheme, easing access to credit for young entrepreneurs.
  output:
    evidence:
      - quote: "to keep our tax regime competitive with neighbouring economies and to encourage private firms to expand investment"
        signal: "Stated goals of tax competitiveness and encouraging private investment — the rationale given for the named corporate rate reduction"
    enacted_signals:
      - quote: "the corporate income tax rate will be reduced by one percentage point, from 25 per cent to 24 per cent"
        signal: "Announced rate reduction for the named measure"
```

```
        timing_signals:
          - quote: "with effect from the 2016 year of assessment"
            signal: "Effective date — 2016 year of assessment"
         # NOT extracted, and why: the consumption-tax rationales ("broadens
the revenue
        # base and strengthens the fiscal position", "reduce the deficit and
achieve a
        # balanced budget by 2020", the exemption-list feedback) all belong to a
        # different measure in the same chunk. The chunk does not tie the deficit
        # framing to the corporate rate reduction, so it is not evidence about
it. The
        # small-enterprise financing allocation is a third, unrelated measure.

  output_instructions: >
    Extract all motivation evidence, enacted-status signals, and timing signals
from
    the chunk that bear on the measure named in the `Act:` field. Evidence
about other
    measures discussed in the same chunk is out of scope and must not be returned.

    For each piece of motivation evidence, provide a direct quote and a brief signal
    description. For each enacted-status signal, provide a direct quote and
a brief
    description of the legislative status it indicates. For each timing signal,
provide
    a direct quote and a brief description of the date and the quarter it
implies under
    Romer & Romer's midpoint rule (when calculable from the quote alone).

    The same direct quote may appear in more than one array if it carries
more than
    one signal type (e.g., a signing date is commonly both an enacted-status signal
    and a fallback timing signal).

    If the chunk contains no evidence of a given type, return an empty array
for that
    field. Empty arrays are valid; do not invent evidence.

    Return your answer as JSON:

    {
      "evidence": [
        {
```

```
          "quote": "Direct quote from the text",
          "signal": "Brief description of the evidence type (e.g., 'stated goal of deficit reduction', 'response to rising unemployment')"
        }
      ],
      "enacted_signals": [
        {
          "quote": "Direct quote about legislative status",
          "signal": "Brief description (e.g., 'signed into law', 'defeated in committee', 'proposed but not passed')"
        }
      ],
      "timing_signals": [
        {
          "quote": "Direct quote about a date or phased implementation",
          "signal": "Brief description (e.g., 'effective date — January 1, 1987 (1987Q1)', 'signing date as fallback timing', 'phased step — July 1, 1982 (1982Q3)')"
        }
      ]
    }
```

C.3 Motivation classification

```
codebook:
  name: "C2b: Motivation Classification (Act-Level)"
  version: "0.9.1"
  # Classification scheme is grounded in Romer & Romer (2010, AER 100(3):
  # 763-801), Section II.C ("Identifying Motivation"), pp. 769-771; and the
  # companion paper "A Narrative Analysis of Postwar Tax Changes," Section I.C
  # ("Classifying Motivation"), pp. 5-8. The four categories, the
  # countercyclical-vs-long-run test, the spending-driven-vs-deficit-driven
  # one-year rule, and the mixed-motivation aggregation rule originate there.
  # The codebook body itself is written in country-agnostic language: author
  # attributions and US-decade-specific anchoring examples have been removed
  # from the prompt the model sees, to avoid US-anchoring the classifier
  # when deployed to Malaysia and other countries (v0.9.1 de-attribution).

  instructions: >
    You are an expert in fiscal policy analysis. You are given consolidated
    act-level evidence about a single fiscal action — a legislated or executive
    measure that changes (or would change) fiscal liabilities or obligations.
    The evidence has already been extracted from contemporaneous primary
```

```yaml
  documents (executive economic outlooks, legislative committee reports,
  head-of-government statements and signing remarks, fiscal authority
  reports). Your task is to classify the **motivation** for the action into
  exactly one of four categories: one of two endogenous categories
  (SPENDING_DRIVEN, COUNTERCYCLICAL) or one of two exogenous categories
  (DEFICIT_DRIVEN, LONG_RUN).

  The central question is: **why did policymakers take the action they
  did?** The framework separates legislated fiscal changes into those taken
  *in response to other factors likely to affect output growth in the near
  future* (endogenous) and those taken *for any other reason* (exogenous).
  Most acts have a single predominant motivation that is consistently
  mentioned across executive and legislative sources. Read the evidence at
  face value: take policymakers' stated reasons as the primary signal of
  motivation, while applying the boundary tests below to distinguish
  neighbouring categories.

  The evidence you receive will typically include three arrays produced by an
  upstream extractor: motivation evidence (direct quotes about *why* the
  action was taken), enacted-status signals (whether the act became law), and
  timing signals (effective dates, signing dates, phased implementation).
  Use the motivation evidence as your primary input for class assignment.
  Use enacted-status signals to fill the `enacted` output field. Use timing
  signals only as supporting context for the spending-driven vs.
  deficit-driven temporal test (see SPENDING_DRIVEN below); do not use them
  to assign a quarter — quarter assignment is a downstream codebook.

  The action may originate in any country and any period. Where the
  evidence references country-specific institutions (legislative committees,
  economic outlook documents, fiscal authorities, social-insurance trust
  funds), treat them as functional equivalents within the framework. The
  underlying *concepts* — return growth to normal, raise growth above
  normal, inherited deficit, raise potential output, smaller government,
  fairness, improved incentives — apply across countries. Do not require
  the specific institutional vocabulary of any one country.

  Read the class definitions and classify the
  act using exactly one of the four labels.

classes:

  - label: "SPENDING_DRIVEN"
    label_definition: >
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        A fiscal action motivated by a contemporaneous change in government
        spending — a revenue measure taken to pay for, or to offset the
        expansionary effects of, an associated spending increase.

      clarification:
      - "The classic case is a revenue increase taken to finance a contemporaneous spending programme — for example, a levy introduced alongside a new public health, social insurance, or defence programme, where policymakers state explicitly that the revenue is intended to pay for the associated spending. Another is a revenue increase to finance a war or defence build-up. Policymakers in such cases often state explicitly that the revenue measure is intended to pay for, or to offset the expansionary effects of, increased government spending."
        - "Spending-driven actions are endogenous because they are tax actions taken to offset another factor that would tend to move output growth away from normal — the associated spending change is itself a force pushing output away from normal, and the revenue measure is a response to it."
        - "**One-year temporal rule.** A revenue increase taken to pay for a *past* spending increase is classified as SPENDING_DRIVEN if it occurs **within roughly one year** of the associated spending increase, and as DEFICIT_DRIVEN if it occurs **more than one year** after. This rule resolves cases — common in social-insurance legislation — where a single bill schedules an immediate benefit increase together with revenue increases that take effect years later: the immediate-financing portion is spending-driven, and the delayed portion is deficit-driven."
       - "Apply the one-year rule using the timing evidence already in the input. If the evidence shows the spending and the revenue change taking effect close together (within about a year), classify as SPENDING_DRIVEN. If the evidence shows the revenue change taking effect well after the associated spending change (more than about a year), the change has shifted into the DEFICIT_DRIVEN category and should be classified there instead."
       - "When the evidence describes a deliberate financing link — phrases such as 'to pay for', 'to finance', 'in order to fund', 'to offset the cost of', or descriptions of the revenue measure as the 'financing' or 'pay-for' element of a programme — that is direct evidence of a spending-driven motivation, provided the temporal rule is satisfied."

      negative_clarification:
       - "Calling a spending programme 'structural', 'long-run', 'permanent', or 'an investment in the future' (e.g., infrastructure, social insurance, education, public investment) does NOT by itself make the financing revenue measure exogenous. If the revenue decision was taken in response to that spending, and the temporal connection is within about a year, the revenue measure is SPENDING_DRIVEN regardless of how the spending programme is framed. The motivation for the
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*revenue* action is the contemporaneous spending change, not the long-run growth properties of the programme being financed."
        - "An action is NOT SPENDING_DRIVEN merely because the evidence mentions government spending in passing or describes the broader fiscal context. Spending-driven classification requires that the spending change be cited as the *reason* for the revenue action, not merely as background."
        - "If the only spending change cited is one that occurred well in the past (more than about a year before the revenue action), the action is DEFICIT_DRIVEN, not SPENDING_DRIVEN. Use the one-year rule to separate the two."
        - "Revenue cuts paired with contemporaneous spending cuts in a deficit-reduction package are NOT spending-driven; their motivation is deficit reduction (DEFICIT_DRIVEN), not the spending change."

    - label: "COUNTERCYCLICAL"
      label_definition: >
        A fiscal action designed to **return output growth to normal** — taken
        in response to current or prospective economic conditions that are
        expected to push growth below (or above) its normal rate in the near
        future.

      clarification:
        - "The defining test: a countercyclical action is one whose stated goal is to **return growth to normal**, not to **raise growth above normal**. Stimulating activity in the face of a predicted recession, or restraining activity in the face of an overheating economy, are both countercyclical. A classic example is a revenue cut enacted because policymakers were explicit that they were cutting taxes because the economy was predicted to fall further, and they were attempting to mitigate the decline."
        - "Countercyclical actions are endogenous because they are taken to offset developments that would cause output growth to differ from normal. They are systematically correlated with other factors affecting output in the short run."
        - "Two complementary diagnostics indicate countercyclical motivation: (a) policymakers state directly that the action is intended to mitigate a downturn, support a faltering economy, restore growth, check a slide, or — in the opposite direction — to restrain inflation or cool an overheating economy; and (b) policymakers' own forecasts of unemployment or activity imply growth that would otherwise be below (or above) normal in the absence of the action. Take policymakers' own statements about what counts as 'normal' at face value."
        - "Phrases such as 'to support the economy', 'to restore growth', 'to mitigate the decline', 'to check the downward slide', 'to provide stimulus', 'to counteract the recession', 'to combat inflation', or 'to cool an overheating economy' — when offered as the *reason* for the action — are direct evidence of countercyclical motivation. Predictions that unemployment will rise (implying
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below-normal growth) or fall sharply from a low base (implying above-normal growth) are corroborating evidence."

      negative_clarification:
           - "An action is NOT COUNTERCYCLICAL merely because the evidence mentions a recession, high unemployment, slow growth, inflation, or other macroeconomic conditions. Such mentions are background, not motivation. The action is countercyclical only if the *stated reason* for it is to offset those conditions and bring growth back to normal."
        - "If policymakers say the economy is doing fine (or, equivalently, that growth is currently at or near normal) but they want growth to be **faster** in the long run, the action is LONG_RUN, not COUNTERCYCLICAL. To be explicit: a very common tax cut is one in which policymakers say that the economy is doing fine (output is growing normally), but they want output to grow faster than normal — that is long-run, not countercyclical."
            - "An action whose stated rationale is structural — improving incentives, raising potential output, increasing fairness, shrinking government — is LONG_RUN even if the evidence also mentions current weak conditions. The countercyclical-vs-long-run boundary is decided by the *stated motive*, not by whether macroeconomic context is mentioned."
         - "An offsetting action — for example, a revenue cut taken to undo a previously legislated exogenous revenue increase because the economy has weakened — is, by the offsetting-changes convention, classified with the **same motivation as the original action it counteracts**, not as countercyclical. This avoids identifying two opposite-motivation actions in a quarter when liabilities did not in fact change."

    - label: "DEFICIT_DRIVEN"
      label_definition: >
        A fiscal action — typically a revenue increase, sometimes coupled with
        spending cuts — designed to reduce an **inherited budget deficit**
        resulting from past economic conditions and past policy decisions,
        rather than from any contemporaneous spending change or current
        cyclical conditions.

      clarification:
         - "The defining criterion: a deficit-driven action is a tax increase designed to reduce an **inherited** budget deficit. Such a change is fundamentally different from a spending-driven action because there is no contemporaneous increase in spending; it is taken in spite of, or regardless of, its effects on output in the short run. The deficit being addressed reflects past economic conditions and budgetary decisions, not current conditions or spending changes."
        - "Direct evidence of deficit-driven motivation includes: descriptions
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of the deficit as 'inherited', 'accumulated', 'long-standing', 'structural', or as the result of 'past decisions' or 'past policy'; statements that the action is 'prudent fiscal policy', is intended to 'restore fiscal balance', 'reduce the structural deficit', 'return the budget to sustainability', or to safeguard the long-term solvency of a major programme (for example, restoring 'actuarial soundness' to a social-insurance fund)."
        - "Deficit-driven actions are exogenous because they are not motivated by current or prospective short-run economic conditions and not motivated by a contemporaneous spending change. Note that they may, however, be paired with spending reductions inside the same deficit-reduction package; the package as a whole is still deficit-driven."
        - "**Delayed financing of past spending.** When a single bill calls for both an immediate spending increase and a much-delayed revenue increase to pay for the higher spending, the *delayed* revenue increase is DEFICIT_DRIVEN if it takes effect more than about one year after the spending increase. This situation is common in social-insurance legislation, where benefit increases are often paired with revenue increases scheduled to occur years later. The rationale: by the time the revenue increase takes effect, there is no contemporaneous spending change, only the inherited liability."

      negative_clarification:
        - "An action is NOT DEFICIT_DRIVEN merely because the evidence mentions a deficit, debt, or fiscal imbalance. The deficit must be cited as the *reason* for the action, and it must be characterised as inherited from past decisions or past conditions, not as the consequence of a contemporaneous spending increase that the action is financing."
        - "If the deficit being addressed was created by a contemporaneous spending increase that the action is financing — and the timing rule places the revenue change within about a year of that spending — the action is SPENDING_DRIVEN, not DEFICIT_DRIVEN. The one-year rule is the dividing line."
        - "If the stated motive for the action is to raise long-run growth, improve incentives, increase fairness, or shrink the size of government — even if the action also has favourable deficit consequences — the action is LONG_RUN, not DEFICIT_DRIVEN. Deficit reduction must be the *primary* stated motive for DEFICIT_DRIVEN."
        - "An action taken in response to a current cyclical deficit (i.e., a deficit caused by an ongoing recession that policymakers expect to fade as the economy recovers) is COUNTERCYCLICAL or LONG_RUN, not DEFICIT_DRIVEN. The deficit must be inherited and structural in policymakers' framing, not the temporary product of present conditions."

    - label: "LONG_RUN"
      label_definition: >
```

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        A fiscal action aimed at **raising long-run growth** or pursuing other
        structural goals — improved incentives, greater efficiency, fairness,
        or a philosophical preference for smaller government — rather than at
        responding to current or prospective short-run economic conditions.

      clarification:
        - "The defining characterisation: long-run actions are those whose goal
is to **raise growth above normal** in the long run, or to **raise potential
output**, in contrast to countercyclical actions whose goal is to return growth
to normal. A quintessential exogenous change is a revenue cut motivated by a
belief that lower marginal tax rates will raise output in the long run, where
the goal is to raise normal growth, not to offset shocks acting to reduce growth
relative to normal."
        - "This category encompasses a wide range of motivations united by their
structural, non-cyclical character: a stated belief that the action will raise
the long-run growth rate of potential output; a desire to improve **incentives**
for work, saving, or investment; a desire for greater **efficiency** of the
fiscal system; a desire for greater **fairness** or equity; or a philosophical
preference for **smaller government**, **limited government**, **economic
competitiveness**, or **structural reform**. These all fall under the broad
rubric of tax changes for long-run growth."
        - "Direct evidence of long-run motivation includes phrases such
as 'raise long-run growth', 'raise potential output', 'remove a barrier to
growth', 'improve incentives for work, saving, and investment', 'remove the
drag on the economy', 'reform the system for fairness', 'simplify the system',
'broaden the base', 'shrink the size of government', 'limited government',
'economic competitiveness', 'structural transformation', or characterisations
of the action as a 'fundamental reform' of the fiscal system rather than a
cyclical response."
        - "**Timing-of-decision rule.** An action proposed when the economy was
growing normally remains LONG_RUN even if the economy weakened by the time of
passage, provided the long-run rationale persists in the evidence and was not
displaced by a new countercyclical motivation at passage. An illustrative case
is an act originally proposed when the economy was growing normally on the basis
of a belief in structural reform and a desire to stimulate long-run growth;
this would be classified as exogenous (long-run) even though, by the time it
passed, concerns about a developing recession were also being mentioned. Use
the *prevailing motivation at the time of passage* — but a long-run rationale
that is still present and dominant at passage is not displaced merely because
cyclical conditions have changed in the interim."
        - "Long-run actions are exogenous because they are not motivated by
current or prospective short-run economic conditions; they are taken to alter
the structural properties of the fiscal system or the long-run path of output."
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      negative_clarification:
          - "Mention of contemporaneous macroeconomic conditions — recession, high unemployment, slow growth, inflation — is NOT, by itself, evidence of a countercyclical motive when the stated rationale for the action is structural or long-run. Long-run actions are sometimes enacted during periods of weak activity in which policymakers nevertheless frame the action as motivated by long-run growth, fairness, or improved incentives. Acknowledging current conditions does not by itself imply endogeneity if the stated motive is explicitly non-cyclical."
        - "An action is NOT LONG_RUN merely because the evidence mentions growth. Policymakers commonly say they want to 'stimulate growth' or 'support growth' — the question is *which growth*. If the stated goal is to **return growth to normal** in the face of a downturn, the action is COUNTERCYCLICAL. If the stated goal is to **raise growth above normal** or to raise the **long-run growth rate of potential output**, the action is LONG_RUN."
            - "An action is NOT LONG_RUN if the primary stated motive is to finance a contemporaneous spending increase (SPENDING_DRIVEN) or to reduce an inherited deficit (DEFICIT_DRIVEN), even if a secondary long-run rationale is also mentioned. Use the predominant motive cited consistently across the evidence."
        - "Vague aspirational language about 'a stronger economy', 'prosperity', or 'national strength' that does not specify whether the goal is cyclical or structural is not, on its own, sufficient evidence of LONG_RUN. Look for explicit references to long-run growth, potential output, incentives, efficiency, fairness, or smaller government."

  extra_output_fields:

    - name: "enacted"
      type: "boolean"
      description: >
        TRUE if the consolidated evidence indicates the act was enacted into
        law or otherwise took legal effect (including via executive or
        administrative action). FALSE if the evidence indicates the measure
        was proposed but not passed, was defeated, was withdrawn, or remained
        a proposal at the time of writing. Use the enacted_signals field of
        the upstream evidence as the primary source. If the enacted-status
        evidence is genuinely silent or ambiguous, default to TRUE only when
        the broader narrative treats the act as a fait accompli (signing
        date stated, retrospective discussion of effects, etc.); otherwise
        default to FALSE.

    - name: "sign"
```

```yaml
    type: "enum"
    values: ["increase", "decrease", "no_change"]
    description: >
      The net direction of the action's effect on fiscal liabilities or
      obligations. Use 'increase' for a net rise in fiscal liabilities (a
      revenue-raising measure such as a rate increase, base broadening, or
      new levy). Use 'decrease' for a net reduction in fiscal liabilities
      (a revenue-reducing measure such as a rate cut, new credit, or
      exemption). Use 'no_change' only when the evidence describes the
      action as net-neutral overall (e.g., a revenue-neutral reform that
      rebalances within the system without raising or lowering aggregate
      liabilities). For phased actions, report the net direction of the
      full set of phases described in the evidence.

  - name: "confidence"
    type: "enum"
    values: ["low", "medium", "high"]
    description: >
      Your confidence in the motivation classification. Use 'high' when
      the evidence consistently and unambiguously cites a single
      category. Use 'medium' when the evidence points predominantly
      to one category but contains some signals consistent with a
      neighbouring category, or when one of the boundary tests
      (countercyclical vs. long-run; spending-driven vs. deficit-driven)
      was needed to resolve the case. Use 'low' when the evidence is
      sparse, internally conflicting, or when assignment required
      substantial judgement beyond what the evidence directly supports.

  - name: "reasoning"
    type: "string"
    description: >
      A brief explanation (2-4 sentences) that names the criterion that
      was decisive for the classification. Cite specific evidence from
      the input — direct quotes or paraphrases of the motivation
      evidence — and explicitly identify which test or clarification was
      applied (for example: 'applies the countercyclical-vs-long-run
      test: stated motive is to raise potential output despite a
      recession context'; or 'applies the one-year temporal rule:
      revenue increase takes effect within a year of the associated
      spending increase, so SPENDING_DRIVEN rather than DEFICIT_DRIVEN';
      or 'applies the offsetting-changes convention: this revenue cut
      counteracts a previously legislated long-run cut and inherits its
      motivation'). Do not write a free narrative; cite the criterion.
```

```
  output_instructions: >
    Classify the act using exactly one of the four motivation labels:
    SPENDING_DRIVEN, COUNTERCYCLICAL, DEFICIT_DRIVEN, or LONG_RUN. Use
    exactly one label; do not output multiple labels, and do not invent
    additional categories. If the evidence genuinely supports more than
    one motivation, assign the motivation that accounts for the largest
    share of the action's fiscal effect, as stated by policymakers, across
    the evidence at the time of passage.

    Apply the four boundary tests embedded in the class definitions:

    - Countercyclical vs. long-run: is the stated goal to return growth to
      normal (countercyclical) or to raise growth above normal / raise
      potential output (long-run)?

    - Spending-driven vs. deficit-driven: is the revenue change paired with
      a contemporaneous spending change within about a year
      (spending-driven), or with a past spending change more than a year
      earlier, or with no contemporaneous spending change at all
      (deficit-driven)?

    - Macroeconomic context vs. countercyclical motivation: mention of
      current conditions is background, not motivation; the action is
      countercyclical only if the stated reason is to offset those
      conditions and bring growth back to normal.

    - Structural framing of a spending programme: calling a spending
      programme 'long-run' or 'structural' does not make the financing
      revenue measure exogenous if the temporal connection to the spending
      change is within about a year.

    Then report the four extra fields described above: `enacted`, `sign`,
    `confidence`, and `reasoning`. The `reasoning` field must cite the
    specific criterion that was decisive — do not write a free narrative.

    Return your answer as JSON with exactly the following keys and no
    others:

    {
         "label": "SPENDING_DRIVEN" | "COUNTERCYCLICAL" | "DEFICIT_DRIVEN" |
"LONG_RUN",
      "enacted": true | false,
```

```
      "sign": "increase" | "decrease" | "no_change",
      "confidence": "low" | "medium" | "high",
      "reasoning": "2-4 sentences citing the decisive criterion and the specific
evidence that triggered it"
    }

    Do not include an `exogenous` field; exogeneity is derived downstream
    from the motivation label.
```

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